



GRADIENT PROJECTION METHODS FOR CONTINUAL LEARNING IN HYPERNETWORKS

OVERCOMING CHUNK-INDUCED GRADIENT
PATHOLOGIES VIA PROJECTION
NORMALIZATION, AND THE INTRODUCTION OF
IFOPNG

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Abstract

This thesis addresses the fundamental incompatibility of applying gradient projection-based continual learning (CL) methods to hypernetwork architectures, establishing a novel framework for stable subspace optimization. Traditional projection methods collapse due to a structural-numerical pathology where the architectural necessity of weight-chunking, required to avoid an output parameter explosion, forces the model to be produced in hundreds or thousands of sequential chunks before a single projection step can execute. This repetitive chunk-generation loop accumulates an exceptionally large gradient volume within the memory matrix, blowing up the condition number of the projection correlation matrix to infinity and causing catastrophic inversion failures that the standard regularization parameter λ cannot rescue.

The primary research objective is to eliminate these numerical singularities and systematically evaluate whether combining hard geometric projections on top of task-conditioned hypernetworks and their output-scoped functional regularizers yields a constructive synergy. To achieve this, we introduce *projection-scoped normalization*: gradient columns stored in the memory matrix G are rescaled to unit ℓ_2 norm, and the diagonal empirical Fisher vector is rescaled by its absolute maximum, together stabilizing the metric tensor within the local projection algebra. We also introduce iFOPNG, an inertial variant of Fisher-Orthogonal Projected Natural Gradient Descent (Garg, Kolhe, Peng, & Gopalam, 2026) that replaces the current-task Fisher F_{new} with the combined curvature $F_c = F_{\text{new}} + F_{\text{old}}$, inspired by the parameter-inertia concept of Elastic Weight Consolidation. Methods are benchmarked on sequential Split and Permuted MNIST, Split-CIFAR10 and Split-CIFAR100. The empirical findings reveal that projection normalization completely resolves matrix inversion collapse, and that in the hypernetwork setting all normalized projection

methods outperform plain SGD, with iFOPNG matching the best projection baseline. However, a standalone hypernetwork optimized with Adam and von Oswald’s functional regularizer consistently surpasses all projection variants, indicating that momentum, which projection methods forgo by design, is the decisive factor in compressed generative weight spaces.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

Data Source. The MNIST (owned by LeCun et al.) and CIFAR (CIFAR-10 and CIFAR-100, owned by Krizhevsky et al.) datasets are public, anonymized datasets licensed for research. These datasets form the basis for the Split-MNIST, Permuted MNIST, Split-CIFAR10, and Split-CIFAR100 benchmarks evaluated in this work. No human or animal data was collected. Data was acquired via the `torchvision.datasets` API.

FIGURES. All figures and plots were created entirely by the author using `matplotlib` and `Weights & Biases`.

CODE. The baseline FOPNG framework was adapted from Garg et al. (2026); all other execution infrastructure was implemented by the author (<https://github.com/kelares/HyperFisher>). Libraries used: Python (v3.10), PyTorch (v2.x), Torchvision, Weights & Biases, NumPy, Matplotlib, and Tqdm.

TECHNOLOGY. The manuscript was typeset in $\text{\LaTeX}$ using BibTeX. During the preparation of this work, the author(s) used Gemini and Claude in order to optimize academic phrasing, outline structure, paraphrase transitions, and check grammar. After using these tools/services, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the thesis. AI Citations: Google. (2026). *Gemini* [Large language model]. <https://gemini.google.com>; Anthropic. (2026). *Claude* [Large language model]. <https://claude.ai>.

2 INTRODUCTION

The capacity for continuous, sequential adaptation without the eradication of historical knowledge remains an architectural cornerstone of biological intelligence, yet it presents a fundamental barrier for artificial neural networks. In deep learning paradigms, sequential training across non-stationary data distributions triggers a systemic pathology known as *catastrophic forgetting* (McCloskey & Cohen, 1989). When a network optimizes

its parameters for an incoming task, gradient updates blindly overwrite the weight configurations essential for retaining prior task spaces, shifting the internal representation manifolds away from historical objective minima.

Modern continual learning (CL) literature is conventionally structured into three core algorithmic families: replay-based strategies that leverage episodic data buffers, regularization-based frameworks that penalize parameter deviations, and architectural or parameter-isolation methods (De Lange et al., 2022). Within the meta-learning domain, hypernetworks (Ha, Dai, & Le, 2016) have emerged as an elegant paradigm. Hypernetworks circumvent parameter-space conflicts by utilizing a centralized, lightweight meta-model to generate the target network’s complete parameter weights conditioned on task embeddings, as formalized by Oswald, Henning, Grewe, and Sacramento (2020). Concurrently, within the field of geometric optimization, gradient projection methods, exemplified by the recently proposed Fisher-Orthogonal Projected Natural Gradient Descent (FOPNG) (Garg et al., 2026), explicitly constrain subsequent updates to lie orthogonal to the subspace of parameter directions critical for previous tasks, with the geometry of orthogonality ranging from Euclidean (OGD) to Fisher-Riemannian (FOPNG).

Despite the mathematical elegance of both hypernetworks and gradient projection methods, their structural intersection uncovers a critical and previously unexamined optimization pathology. Generating a target network’s full weight vector in a single forward pass would require an output layer whose width equals the total number of target network parameters, which is computationally prohibitive for any non-trivial architecture; Because every sub-pass executes before any gradient update, their contributions accumulate additively in the shared meta-model parameters, inflating the gradient magnitude proportionally to K . This gradient inflation propagates into the projection algebra: both the gradient columns stored in memory matrix G and the diagonal Fisher estimate $\hat{F}$ grow in scale with K , rendering the regularization term λI negligible relative to the inflated projection matrix $G^\top \hat{F} G$. The condition number of this matrix exceeds 10^{10} , causing catastrophic inversion failures.

Research Objectives and Algorithmic Framework

The primary research objective of this thesis is to systematically resolve this linear-algebraic collapse, map the initialization mechanics of meta-projection pipelines, and evaluate whether the integration of hard geometric constraints on top of soft functional regularizers yields an optimized sequential learning system.

To achieve this, we introduce *projection-scoped normalization* (column-wise ℓ_2 orthonormalization of G combined with absolute-maximum scaling of $\hat{F}$) and iFOPNG, an inertial FOPNG variant using the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$. These contributions enable stable gradient projection in chunked hypernetworks, with direct applicability to on-device sequential learning where retaining adaptive momentum is critical.

Research Questions

To systematically navigate these boundaries, this thesis addresses the following primary research question:

What numerical pathologies arise when gradient projection methods are applied to chunked hypernetwork architectures, and do modifications to the Fisher metric construction—specifically an inertial combined metric and a monotonic accumulation operator—yield principled improvements over standard projection baselines in continual learning?

This is decomposed into four sub-research questions (Sub-RQs):

- Sub-RQ1 (Synergy vs. Conflict):** Does the joint integration of hard gradient projection with the hypernetwork’s soft functional regularizer yield a constructive synergy, or does a standalone regularized hypernetwork offer a more efficient optimization frontier?
- Sub-RQ2 (Gradient Pathology):** How does the architectural necessity of spawning K sequential weight-chunks inflate the gradient and Fisher magnitudes accumulated before a single projection step, and does rescaling both components independently within the projection algebra suffice to prevent matrix inversion collapse?
- Sub-RQ3 (Parameter Inertia):** Does replacing FOPNG’s task-specific metric $\hat{F}_{\text{new}}$ with the combined inertial metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ yield consistent retention improvements across benchmarks, and does the magnitude of any advantage scale with inter-task Fisher overlap?
- Sub-RQ4 (Fisher Accumulation):** Does accumulating $\hat{F}_{\text{old}}$ via a non-decaying element-wise maximum operator maintain long-horizon retention more effectively than an exponential moving average, and does the monotonic growth of the maximum operator impose a measurable plasticity cost over extended task sequences?

Summary of Empirical Findings

Projection-scoped normalization fully resolves the matrix inversion pathology, recovering stable projection performance across all seeds. The inertial combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ consistently improves retention over FOPNG, with the advantage scaling monotonically with inter-task Fisher overlap. The MAX accumulation operator outperforms EMA on long task sequences and requires no additional hyperparameters. In the standard-capacity hypernetwork setting, iFOPNG surpasses plain SGD but falls short of Adam; under extreme compression the gap to Adam widens to 20–40 pp and most projection methods drop below SGD entirely. Two structural limitations bound the current approach: the loss of adaptive momentum at each projection step, which cannot be recovered without projection-compatible optimizers; and the heuristic nature of the normalization scheme, which stabilises the projection algebra by discarding the curvature structure that a geometrically principled rescaling would preserve.

3 RELATED WORK

This section builds the theoretical and empirical context for the thesis, tracing a path from foundational continual learning challenges through regularization and projection methods, to hypernetwork architectures and the structural gap at their intersection.

3.1 Continual Learning and the Stability-Plasticity Dilemma

Modern machine learning systems are predominantly trained on fixed datasets drawn from a single stationary distribution — a condition that rarely holds in deployment, where agents must adapt to non-stationary data streams while retaining prior knowledge. The fundamental tension between these two objectives is formalized as the *stability-plasticity dilemma* (Grossberg, 1980): an agent must remain plastic enough to absorb novel distributions yet stable enough to resist the erasure of established representations. When standard architectures trained by empirical risk minimization are exposed to sequential data, they suffer *catastrophic forgetting* (McCloskey & Cohen, 1989) — abrupt and near-total loss of previously learned capabilities upon exposure to new task data.

The continual learning literature addresses this across three canonical scenarios (De Lange et al., 2022): *task-incremental learning*, where the task identity is available at both training and test time; *class-incremental learning*, where task identity is withheld at inference; and *domain-incremental learning*, where the output space is shared but the input distribution shifts

continuously. This thesis focuses on the task-incremental setting, the scenario in which projection-based methods and hypernetwork architectures have been most thoroughly evaluated and in which structural comparisons between them are most tractable.

3.2 Evaluation Protocols: Multi-Head and Single-Head Designs

Within the task-incremental setting, two evaluation protocols govern how the network’s output layer is organized. In the *multi-head* protocol, each task is assigned a dedicated output head selected at test time using the known task identity, eliminating output-level interference between tasks. In the *single-head* protocol, all tasks share one output layer, requiring the backbone alone to resolve inter-task interference — a substantially harder problem. Farquhar and Gal (2018) demonstrated that multi-head evaluations systematically inflate apparent method performance by removing a major source of forgetting, and argued that single-head evaluation is the more rigorous protocol.

This thesis uses multi-head benchmarks for all Split-CIFAR experiments to match the evaluation conditions of Garg et al. (2026) and enable direct numerical comparison. For Split-MNIST, both protocols are evaluated: the multi-head variant replicates Garg et al.’s conditions, while the single-head variant follows Farquhar and Gal (2018)’s recommendation.

It should be noted that Garg et al. (2026) label their Split-MNIST evaluation as single-head; however, replication of their stated first-task protocol — SGD at $\eta = 10^{-5}$ for 5 epochs — yields approximately 21% first-task accuracy, barely above the random baseline, making their reported results unattainable under the stated conditions. We therefore suspect the original evaluation used multi-head design or a different first-task hyperparameter, and report both protocols to provide the most informative comparison regardless of which interpretation is correct.

3.3 Regularization-Based Approaches

The dominant paradigm for addressing catastrophic forgetting is *parameter regularization*, of which Elastic Weight Consolidation (EWC) (Kirkpatrick et al., 2017) is the canonical representative. EWC estimates the structural importance of each weight using the diagonal of the empirical Fisher Information Matrix $\hat{F}$, and applies a soft quadratic penalty that resists changes to parameters important to prior tasks:

$$\mathcal{L}_{\text{EWC}}(\theta) = \mathcal{L}_t(\theta) + \frac{\lambda}{2} \sum_i \hat{F}_i (\theta_i - \theta_{i,t-1}^*)^2. \quad (1)$$

EWC is computationally efficient and broadly applicable, and serves as the standard reference point against which new continual learning methods are measured. Its principal limitation is that soft penalties permit bounded parameter drift: cumulative updates across a long task sequence gradually erode the protection of earlier task spaces, a failure mode that becomes increasingly pronounced as T grows (De Lange et al., 2022).

3.4 Gradient Projection Methods

To enforce strict, non-negotiable protection for prior knowledge rather than relying on penalties that can be overcome by sufficiently large gradients, a second paradigm uses explicit *gradient projection*. Orthogonal Gradient Descent (OGD) (Farajtabar, Azizan, Mott, & Li, 2020) maintains a memory matrix $G = [g_1 \mid \cdots \mid g_K]$ of historical task gradient directions and projects incoming gradients onto the orthogonal complement of $\text{Col}(G)$ before each update:

$$\tilde{g} = g - G(G^\top G)^{-1}G^\top g. \quad (2)$$

This provides a hard guarantee: the update cannot increase the loss on previously seen tasks. Generalization bounds established within the Neural Tangent Kernel regime confirm that OGD provides exact protection in linear and mildly nonlinear settings (Bennani, Doan, & Sugiyama, 2020).

The mathematical appeal of projection methods over EWC lies precisely in this hardness: where EWC applies a soft penalty that can be overwhelmed, projection methods impose a geometric constraint that is either satisfied or not. However, standard Euclidean projection treats the parameter space as flat and assigns equal importance to all directions. Fisher-Orthogonal Projected Natural Gradient Descent (FOPNG) (Garg et al., 2026) addresses this by performing the projection in the Riemannian geometry defined by the Fisher metric, so that the orthogonality constraint respects the curvature of the loss surface. The related Fisher Natural Gradient (FNG) (Garg et al., 2026) applies the natural gradient without projection, and Orthogonal Natural Gradient (ONG) (Yadav, Mendoza, & Korrapati, 2025) combines natural gradient preconditioning with Euclidean projection — sitting structurally between OGD and FOPNG.

3.5 Hypernetworks for Continual Learning

An orthogonal architectural response to catastrophic forgetting replaces the multi-head output design entirely with a *hypernetwork* (Ha et al., 2016): a meta-model $\mathcal{H}(\cdot; \phi)$ trained to synthesize the complete weight vector $\theta_t = \mathcal{H}(e_t; \phi)$ of a target network, conditioned on a task embedding e_t . Each

task receives a unique weight configuration generated on demand, rather than a dedicated output head, achieving implicit task separation through the generative bottleneck and impressive parameter compression ratios relative to storing separate networks per task. Generating the full θ_t in a single forward pass is computationally intractable for large target networks, so [Oswald et al. \(2020\)](#) introduced *weight chunking*: the generator iterates over a secondary chunk embedding c_k , producing C_s target parameters per pass and assembling the full weight vector by concatenation. To prevent the hypernetwork from forgetting how to generate prior task configurations, a soft *functional regularizer* is applied in the output space of the generator:

$$\mathcal{L}_{\text{reg}} = \beta \sum_{s=1}^{t-1} \|\mathcal{H}(\phi, e_s) - \theta_s^*\|_2^2, \quad (3)$$

where θ_s^* denotes the weight snapshot saved after task s . This penalty can be interpreted as EWC applied at the meta level: rather than anchoring individual network parameters, it anchors the generated weight configurations for each previously seen task, penalizing drift in the space where task-relevant quantities actually live. Recent extensions have refined the hypernetwork approach: Partial Hypernetworks ([Hemati, Lomonaco, Bacciu, & Borth, 2023](#)) reduce the computational overhead of full weight generation, and Prototype Augmented Hypernetworks ([Fuente et al., 2025](#)) augment task embeddings with prototype representations to further stabilize sequential learning.

3.6 Identified Gaps and Motivating Pathologies

Whether hard geometric projection constraints are compatible with the soft functional regularization already employed by hypernetworks, or whether layering both simultaneously over-constrains the generator’s plasticity, has not been investigated (Sub-RQ1).

The architectural necessity of weight-chunking introduces a gradient accumulation pathology — inflated condition numbers in the projection kernel — that is unique to the hypernetwork and projection methods combination and has no prior treatment in the literature (Sub-RQ2).

Existing projection methods enforce a rigid subspace boundary on historical gradient directions; whether replacing this boundary with a combined Fisher metric encoding both old and new task curvature yields consistent retention improvements across benchmarks has not been examined (Sub-RQ3).

The historical Fisher estimate underlying Fisher-based projection depends on a choice of accumulation operator; whether a monotonic max-

imum or a decaying exponential moving average better preserves long-horizon retention has not been directly compared (Sub-RQ4).

These four gaps jointly motivate the sub-research questions stated in Section 2 and the experimental programme described in Section 4.6.

4 METHODS

This section specifies the precise formulation of every method and architectural component used in our experiments, providing sufficient detail for reproducibility. Conceptual motivation and positioning relative to prior work are covered in Section 3; we restrict our attention here to the exact algorithmic instantiations evaluated.

4.1 Continual Learning Protocol

4.1.1 Task Setup and Evaluation Metrics

A continual learning agent receives a sequence of T tasks $\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_T$ presented in order. At task t , only the data from $\mathcal{T}_t$ is available; prior task data is not retained. After each task is trained, the model is evaluated on all tasks seen so far. We report two aggregate metrics: *average accuracy*

$$A_t = \frac{1}{t} \sum_{i=1}^t a_{t,i}, \quad (4)$$

and *backward transfer*

$$\text{BWT} = \frac{1}{T-1} \sum_{t=2}^T (a_{T,t} - a_{t,t}), \quad (5)$$

where $a_{i,j}$ denotes accuracy on task j immediately after training task i . Negative BWT quantifies catastrophic forgetting; a value of zero indicates perfect retention.

4.1.2 Multi-Head and Single-Head Evaluation

Following Farquhar and Gal (2018), who demonstrated that multi-head evaluations can inflate apparent method performance, we report results under both protocols for MNIST-based benchmarks.

4.2 Fisher Information Matrix and Regularization-Based Continual Learning

4.2.1 Fisher Information Matrix

All Fisher-based methods in this thesis use the *diagonal empirical approximation*

$$\hat{F}_i \approx \frac{1}{N} \sum_{n=1}^N \left(\frac{\partial \mathcal{L}^{(n)}}{\partial \theta_i} \right)^2, \quad (6)$$

estimated over N held-out samples at the end of each task. The full FIM is intractable for our network sizes; the diagonal approximation is standard across all evaluated methods (Garg et al., 2026; Kirkpatrick et al., 2017).

4.2.2 Elastic Weight Consolidation

EWC (Kirkpatrick et al., 2017) augments the task loss with a Fisher-weighted quadratic penalty:

$$\mathcal{L}_{\text{EWC}}(\theta) = \mathcal{L}_t(\theta) + \frac{\lambda}{2} \sum_i \hat{F}_i (\theta_i - \theta_{i,t-1}^*)^2, \quad (7)$$

where θ_{t-1}^* denotes the parameter snapshot after task $t-1$. EWC operates exclusively through parameter-space penalties and maintains no gradient subspace; it is included as a regularization baseline employing a different mechanism from projection. We use $\lambda = 10^{-3}$ for all hypernetwork configurations and reproduce the per-benchmark values from Garg et al. (2026) Table 1 for standalone evaluations. Fisher estimates are computed over the full training set for MNIST-based benchmarks and over $N=1024$ samples for CIFAR benchmarks.

4.3 Gradient Projection Methods

This subsection presents the full suite of gradient-based methods evaluated in this thesis. SGD and FNG establish unconstrained lower bounds with no forgetting-protection mechanism; OGD through iFOPNG form the projection family that is the primary focus of this work.

4.3.1 Stochastic Gradient Descent

SGD applies vanilla gradient descent with no forgetting protection:

$$\Delta \theta_{\text{SGD}} = -\eta \nabla_{\theta} \mathcal{L}_t. \quad (8)$$

Adam (?) is included as an adaptive-rate variant. Both define the lower bound on retention performance; any continual learning method should improve upon SGD on the forgetting axis.

4.3.2 Natural Gradient and Fisher Natural Gradient

Natural gradient descent (Amari, 1998) preconditions the gradient with the inverse FIM, respecting the curvature of the statistical manifold:

$$\Delta\theta_{\text{nat}} = -\eta \hat{F}^{-1} \nabla_{\theta} \mathcal{L}. \quad (9)$$

With a diagonal Fisher approximation, inversion reduces to element-wise scaling and remains computationally tractable.

FNG (Garg et al., 2026) applies a trust-region constraint to this Riemannian step:

$$v_{\text{FNG}}^* = -\eta \frac{\hat{F}^{-1} g}{\sqrt{g^{\top} \hat{F}^{-1} g}}. \quad (10)$$

FNG imposes no projection subspace and constitutes no continual learning mechanism; it is included to isolate the contribution of Fisher preconditioning from that of gradient projection.

4.3.3 Orthogonal Gradient Descent

OGD (Farajtabar et al., 2020) maintains a memory matrix $G = [g_1 \mid \cdots \mid g_K] \in \mathbb{R}^{p \times K}$ of past task gradient directions and updates parameters via the current gradient projected onto the orthogonal complement of $\text{Col}(G)$:

$$\tilde{g} = g - G(G^{\top} G)^{-1} G^{\top} g, \quad \Delta\theta_{\text{OGD}} = -\eta \tilde{g}. \quad (11)$$

OGD operates in Euclidean space, treating all parameter directions symmetrically.

GRADIENT MEMORY MANAGEMENT. In all projection methods, G is maintained as a fixed-budget basis of at most M columns. Garg et al. (2026) set $M = T \times K$ across all benchmarks, so compression never triggers in their experiments. We address this unexamined regime explicitly: incoming gradient batches are appended directly to G ; when the column count exceeds M , the full basis is recompressed via truncated SVD and $G \leftarrow U_{:,1:M}$, retaining the M directions of greatest variance in the accumulated gradient memory. When projection-scoped normalization is enabled, QR decomposition is additionally applied to each incoming batch at insertion time (Section 4.5.1).

4.3.4 Orthogonal Natural Gradient

ONG applies OGD-style projection to the natural gradient rather than to the raw gradient, combining Fisher preconditioning with a Euclidean orthogonality constraint:

$$\Delta\theta_{\text{ONG}} = -\eta \left(\hat{F}^{-1} g - G(G^{\top} G)^{-1} G^{\top} \hat{F}^{-1} g \right). \quad (12)$$

Unlike FOPNG, the projection is measured in Euclidean rather than Fisher metric space, so the orthogonality guarantee does not extend to the statistical manifold. ONG uses the same memory budget as OGD.

4.3.5 Fisher-Orthogonal Projected Natural Gradient Descent

FOPNG (Garg et al., 2026) executes projection in the Riemannian metric defined by the Fisher:

$$v_{\text{FOPNG}}^* = -\eta \frac{\hat{F}^{-1}Pg}{\sqrt{(Pg)^\top \hat{F}^{-1}Pg}}, \quad P = I - \hat{F}G \left(G^\top \hat{F}G + \lambda I \right)^{-1} G^\top, \quad (13)$$

where λ is added to the diagonal for numerical stability. We reproduce the per-benchmark learning rates and λ values from Garg et al. (2026) Table 1 for standalone configurations, and use a uniform $\text{lr}=10^{-3}$, $\lambda=10^{-3}$ for all hypernetwork configurations.

4.3.6 Inertial FOPNG

In FOPNG, the parameter update takes the form

$$v = c \hat{F}_{\text{new}}^{-1}(g - u), \quad (14)$$

where $\hat{F}_{\text{new}}$ serves as the Riemannian metric: both the trust-region constraint $\|v\|_{\hat{F}_{\text{new}}} = \eta$ and the projection objective $\|g - \hat{F}_{\text{new}}^{-1}u\|_{\hat{F}_{\text{new}}}^2$ are measured in the current task’s Fisher geometry. This is natural when no prior tasks exist, but creates an asymmetry once history accumulates: the constraint subspace $\text{Col}(G)$ is built from gradient directions collected under $\hat{F}_{\text{old}}$, yet the projection metric assigns zero curvature to parameters that were important in previous tasks but uninformative on the current one.

iFOPNG is introduced in this thesis as a variant of FOPNG that resolves this asymmetry by adopting the *combined metric*

$$F_c \triangleq \hat{F}_{\text{new}} + \hat{F}_{\text{old}}, \quad (15)$$

where $\hat{F}_{\text{old}}$ is maintained as the element-wise maximum of all per-task Fisher diagonals accumulated so far. Replacing (14) with $v = c F_c^{-1}(g - u)$, the constrained trust-region update becomes

$$v^* = -\eta \frac{F_c^{-1}Pg}{\sqrt{(Pg)^\top F_c^{-1}Pg}}, \quad P = I - \hat{F}_{\text{old}} G A^{-1} G^\top \hat{F}_{\text{old}}, \quad (16)$$

where $A = G^\top \hat{F}_{\text{old}} F_c^{-1} \hat{F}_{\text{old}} G$. A formal derivation is provided in Appendix A.

The combined metric F_c assigns each coordinate i an effective mass $(\hat{F}_{\text{new}})_i + (\hat{F}_{\text{old}})_i$, suppressing the natural-gradient step size in directions important to any prior task. We call this property *parameter inertia*: accumulated Fisher importance makes a coordinate harder to displace without any explicit penalty term. In direct contrast to EWC (Kirkpatrick et al., 2017), which pulls parameters toward a stored reference point θ^* , iFOPNG maintains no reference point: a parameter that mattered before is not pulled back, it is simply made harder to displace.

4.4 Hypernetwork Architecture

4.4.1 Chunked Hypernetworks

A hypernetwork (Ha et al., 2016) is a meta-model $\mathcal{H}(\cdot; \phi)$ that generates the complete weight vector $\theta_t = \mathcal{H}(e_t; \phi)$ of a target network conditioned on a task embedding e_t . Generating θ_t in a single forward pass would require an output layer of dimension $|\theta_t|$, which is computationally prohibitive for any non-trivial target architecture. *Weight chunking* (Oswald et al., 2020) resolves this by iterating over a secondary chunk embedding c_k , $k = 1, \dots, K$, and producing a block of C_s target parameters per pass; the full weight vector is assembled by concatenation. Chunking makes the hypernetwork parameter-efficient but introduces the chunk-induced gradient accumulation pathology addressed in Section 4.5.1.

4.4.2 Functional Regularizer

The functional regularizer of Oswald et al. (2020) is active for all hypernetwork configurations, with regularization strength β treated as a tunable hyperparameter (see Appendix 8).

4.4.3 Target Network and Dropout

Standalone configurations reproduce the target network architecture of Garg et al. (2026) without modification, including two Dropout($p=0.5$) layers in the SimpleCIFARCNN classifier head. Dropout is removed from the target network in all hypernetwork configurations: PyTorch’s `functional_call` propagates the parent module’s `.training` mode to the target submodule, causing stochastic masking across all chunk-generation passes during training. The generative bottleneck ($\phi : d_t + d_c \rightarrow d_h \rightarrow C_s$) and the functional regularizer $\mathcal{L}_{\text{reg}}$ provide sufficient implicit regularization in its place.

4.5 Novel Algorithmic Contributions

4.5.1 Projection-Scoped Normalization

When a hypernetwork spawns K sequential weight-chunks before a single projection step executes, the accumulated gradient vector and the diagonal Fisher entries grow in magnitude proportionally to K , inflating the condition number of the projection kernel $G^\top \hat{F} G$ and causing the matrix inversion in Equation (13) to fail catastrophically for any practically sized target network.

We resolve this with two independent rescaling operations that address distinct sources of ill-conditioning:

1. **Gradient basis orthonormalization.** Each batch of K gradient vectors collected after task t is replaced by the orthonormal factor Q from its thin QR decomposition before insertion into G :

$$[g_{t,1} \mid \cdots \mid g_{t,K}] \xrightarrow{\text{QR}} Q, \quad Q^\top Q = I. \quad (17)$$

QR is chosen over SVD at insertion time because the operation requires only orthonormalization of the incoming batch — the number of columns is preserved, not reduced, so the ordered factorization that SVD provides is unnecessary overhead. SVD is reserved for the overflow handler: when the column count of G exceeds the budget M , the full basis is recompressed as $G \leftarrow U_{:,1:M}$, retaining the M left singular vectors of greatest variance and discarding the rest. This operation applies *at storage time* and affects only the basis G ; the gradient g entering the projection update itself is the unmodified parameter gradient from the model.

2. **Metric-tensor normalization.** Within each projection computation, both Fisher terms are divided by the maximum entry of the metric tensor F_c that defines the constrained natural gradient problem:

$$\tilde{F} = \frac{F}{\max_i (F_c)_i + \epsilon'}, \quad F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}, \quad (18)$$

where F_c is the combined Fisher used as the Riemannian metric in iFOPNG, or $\hat{F}_{\text{new}}$ alone for FOPNG. This scaling is applied locally within the kernel computation and does not affect the optimisation trajectory outside the projection step.

Together, these operations ensure $G^\top \tilde{F} G$ remains well-conditioned regardless of the number of chunks spawned per task. Gradient orthonormalization eliminates column-scale inflation in G ; metric-tensor normalization

bounds the Fisher entries to $[0, 1]$, preventing the kernel from becoming numerically singular. Subspace membership is invariant to positive scalar rescaling of basis vectors, so neither operation alters the geometry of the projection.

4.6 Experimental Design

The four sub-research questions demand qualitatively different experimental designs. Sub-RQ1 and Sub-RQ3 require head-to-head benchmark comparisons — projection methods against an unconstrained baseline, and iFOPNG against FOPNG respectively. Sub-RQ2 requires a controlled ablation over normalization conditions. Sub-RQ4 requires a long-horizon comparison of two Fisher accumulation operators on the same elastic preconditioner. Hyperparameter choices fall into two categories: those inherited directly from the literature, which require no further justification, and those specific to the novel hypernetwork experiments, each of which is justified below.

4.6.1 Hyperparameter Provenance

STANDALONE TARGET-NETWORK EXPERIMENTS. All hyperparameter values for experiments without a hypernetwork — learning rates, regularization coefficients, gradient budgets, Fisher sample counts, batch sizes, and epoch counts — are taken without modification from Garg et al. (2026) Table 1. iFOPNG uses the same values as FOPNG, since the inertial combined Fisher introduces no hyperparameters beyond those FOPNG already defines. ONG (Yadav et al., 2025) is treated as a composite method: it applies OGD’s Euclidean projection operator to the natural gradient direction and introduces no hyperparameters beyond those already specified for OGD and FNG; its values therefore follow OGD’s Table 1 entries.

DEVIATION FOR FIRST-TASK INITIALIZATION. We deviated from the original protocol in one specific instance: first-task initialization for MNIST-based benchmarks. While Garg et al. (2026) state they used the method’s own learning rate for first-task SGD training (e.g., $\eta = 10^{-5}$ for FOPNG), our empirical replication of this protocol yielded an initial accuracy of only $\approx 21\%$. As demonstrated by the learning rate convergence sweep detailed in Appendix C (Figure 15), this published rate falls well below the empirical convergence threshold, effectively failing to establish a functional baseline. To ensure a clean initial projection subspace, we standardized the first-task SGD learning rate to 10^{-3} for MNIST-based benchmarks and 10^{-2}

for CIFAR-based benchmarks. Beyond this necessary empirical correction, no tuning was performed on the published values.

HYPERNETWORK EXPERIMENTS. Settings not covered by Garg et al. (2026) follow the canonical hypernetwork framework of Oswald et al. (2020); exact values are detailed in Appendix 8. Because computing a full Fisher estimate requires a complete K -chunk generation pass per training sample, estimation is capped at 1,024 samples to maintain computational tractability — a standard approximation consistent with prior CIFAR-10 evaluations (Kirkpatrick et al., 2017). The gradient budget remains $K = 80$ per task, with memory compression handled via QR-and-SVD (Section 4.3). First-task training uses AdamW to seed the memory matrix G and Fisher estimate $\hat{F}$ from a clean geometric basis.

UNIFORM LEARNING RATE CONTROL. To isolate the projection constraint as the sole independent variable in the hypernetwork experiments, all methods use a uniform learning rate of 10^{-3} rather than method-specific rates. This prevents convergence-speed artifacts from conflating the structural effects of the projection geometry..

4.6.2 Sub-RQ1 — Constructive Synergy or Architectural Conflict

EXPERIMENTAL DESIGN. Sub-RQ1 investigates whether layering a hard geometric projection on top of a hypernetwork’s soft functional regularizer yields a net benefit, or if the loss of adaptive momentum makes a standard Adam-regularized hypernetwork superior. To separate architectural conflicts from benchmark-specific artifacts, we evaluate across three datasets at varying levels of compression, comparing an unconstrained standalone network (Panel a) against a hypernetwork (Panel b).

BENCHMARK SELECTION. We utilize a three-benchmark progression. **Benchmark 1 (Split-MNIST)** utilizes a deliberately “suffocated” hypernetwork (generator hidden dimension $d_h = 8$) to expose differential effects without hitting the near-perfect accuracy ceilings observed at higher capacities ((see Appendix 8, Figure 10)). To ensure convergence under these constraints, each task is trained for 15 epochs with a batch size of 64. **Benchmark 2 (Split-CIFAR10)** serves as the primary test, offering genuine inter-task interference and utilizing a standard-capacity hypernetwork trained for 50 epochs. **Benchmark 3 (Split-CIFAR100)** extends this sequence to 10 tasks to test scalability. Comparing performance within and across these panels reveals whether projection methods universally dominate, conditionally succeed based on compression, or systematically fail due to momentum stripping.

4.6.3 Sub-RQ2 — Projection-Scoped Normalization

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ2 asks whether the chunk-induced gradient accumulation pathology is real, characterisable, and resolvable. A single accuracy number cannot answer this: we need direct numerical evidence that the pathology occurs without normalization, that normalization resolves it, and that the resolution does not compromise projection quality.

EXPERIMENTAL DESIGN. iFOPNG is run on the Split-CIFAR10 hypernetwork (Benchmark 2) under three normalization conditions, all other hyperparameters held constant:

CONDITION 1 — NO NORMALIZATION (NEGATIVE CONTROL). Incoming gradient batches are appended to G without orthonormalization; the Fisher diagonal is used at its raw scale. With several hundred chunks accumulated per forward pass, the gradient columns of G and the entries of $\hat{F}$ are expected to reach extreme magnitudes, inflating the condition number of the projection kernel and driving the matrix inversion to NaN. This condition is the negative control that demonstrates the pathology concretely.

CONDITION 2 — GRADIENT ORTHONORMALIZATION ONLY. Each incoming gradient batch is orthonormalized via QR decomposition before insertion into G ; the Fisher diagonal is used at its raw scale. This isolates whether gradient-side conditioning alone is sufficient, or whether Fisher-side scaling is also required.

CONDITION 3 — FULL NORMALIZATION (PROPOSED). Gradient batches are QR-orthonormalized at insertion, and the Fisher terms are scaled by the maximum entry of the metric tensor F_c within each projection computation. This is the proposed scheme; all other hypernetwork experiments use it.

In addition to accuracy and BWT, the condition number of the projection kernel A is logged before each inversion. A condition number that diverges in Condition 1 but remains bounded in Condition 3 constitutes direct numerical evidence for the pathology and its resolution. All conditions use the Benchmark 2 hypernetwork settings and the uniform learning rate of 10^{-3} , so any performance difference is attributable to normalization alone.

4.6.4 Sub-RQ3 — Does iFOPNG’s Elastic Preconditioner Help

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ3 asks whether the elastic combined Fisher $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ provides a consistent retention improve-

ment over rigid FOPNG, and whether any advantage is robust to the evaluation protocol.

BENCHMARK SELECTION AND RATIONALE. To test the robustness of the elastic preconditioner, the full method suite is evaluated across four standalone benchmarks. **Permuted-MNIST** tests long-horizon retention under high inter-task Fisher overlap. **Split-MNIST** is evaluated under both multi-head (Garg et al., 2026) and single-head (Farquhar & Gal, 2018) protocols to determine if the iFOPNG advantage survives output-level interference. Finally, **Split-CIFAR10** and **Split-CIFAR100 (multi-head)** provide the most discriminative environments with stronger inter-task interference. Hypernetwork configurations for Split-MNIST and Split-CIFAR10 are also included for completeness to verify whether the elastic advantage survives extreme parameter compression. Comparing iFOPNG against FOPNG across this suite isolates the specific retention benefit of the elastic preconditioner, while comparisons against OGD verify the necessity of the Fisher metric itself.

4.6.5 Sub-RQ4 — Fisher Accumulation Operator

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ4 holds the elastic formulation of Sub-RQ3 fixed and isolates the operator used to accumulate the historical Fisher F_{old} . The non-decaying maximum $F_{\text{old}} \leftarrow \max(F_{\text{old}}, F_{\text{new}})$ guarantees that any parameter ever deemed important to a past task retains that importance permanently. This is protective for early tasks but monotonic by construction: the elastic inertia boundary can only tighten, never relax, so in a sufficiently long sequence the accumulated metric may immobilise the parameters needed to learn later tasks.

EXPERIMENTAL DESIGN. iFOPNG is run under two accumulation operators, all other settings held constant:

MAXIMUM (PROPOSED). $F_{\text{old}} \leftarrow \max(F_{\text{old}}, F_{\text{new}})$ — non-decaying.

EXPONENTIAL MOVING AVERAGE. $F_{\text{old}} \leftarrow (1 - \alpha) F_{\text{old}} + \alpha F_{\text{new}}$ — decaying, allowing historical importance to fade.

The comparison is run on the longest available task sequences — Permuted-MNIST — since the plasticity cost of monotonic accumulation, if present, only becomes observable as the sequence length grows. Per-task accuracy is tracked across the full sequence rather than reported only as a final aggregate: the diagnostic signature of plasticity paralysis is a progressive decline in *newly learned* task accuracy ($a_{t,t}$ falling as t increases), distinct from forgetting, which appears as decline in *previously learned* task accuracy.

If the maximum operator shows superior early-task retention but degrading $a_{t,t}$ in late tasks, while the moving average shows the opposite trade, the two failure modes are cleanly separated and Sub-RQ4 is answered in full.

5 RESULTS

This section reports the empirical outcomes of the four sub-research questions in the order stated in the introduction. For each, we describe the experimental scope, present the key quantitative findings, and highlight the most important patterns; detailed interpretation is reserved for the Discussion. *Average accuracy* (A_T) is the primary evaluation metric, and *backward transfer* (BWT) is reported as a secondary measure of forgetting. Standalone target-network results that establish the reference behaviour of projection methods in their original, non-hypernetwork setting are provided in Appendix B (Figures 7–9).

5.1 Sub-RQ1: Constructive Synergy or Architectural Conflict?

This subsection examines whether combining gradient projection with the hypernetwork’s functional regularizer yields better continual learning than either component alone. Three hypernetwork regimes are evaluated: a standard-capacity Split-CIFAR10 network, an extremely compressed Split-MNIST network (suffocated regime, $d_h = 8$), and a ten-task Split-CIFAR100 network.

STANDARD HYPERNETWORK — SPLIT-CIFAR10. Figure 1 presents average accuracy and BWT for all eight methods on Split-CIFAR10 with a standard-capacity hypernetwork and full normalization enabled (see Section 5.2). Plain SGD at 84.1% establishes the unconstrained lower reference point, and Adam at 90.1% defines the upper frontier a standalone hypernetwork can reach without projection. Every projection method surpasses SGD: iFOPNG leads at 85.6%, followed by EWC (81.5%), FNG (80.9%), OGD (78.3%), FOPNG (77.5%), and ONG (67.2%), confirming a partial constructive synergy with the hypernetwork’s regularizer. However, none match Adam, which leads by 4.5 pp over the best projection method.

The BWT values reveal an instructive asymmetry: Adam has effectively zero backward transfer ($\text{BWT} \approx 0.000$), while all projection methods exhibit small but nonzero forgetting (iFOPNG: -0.013 ; FOPNG: -0.015).

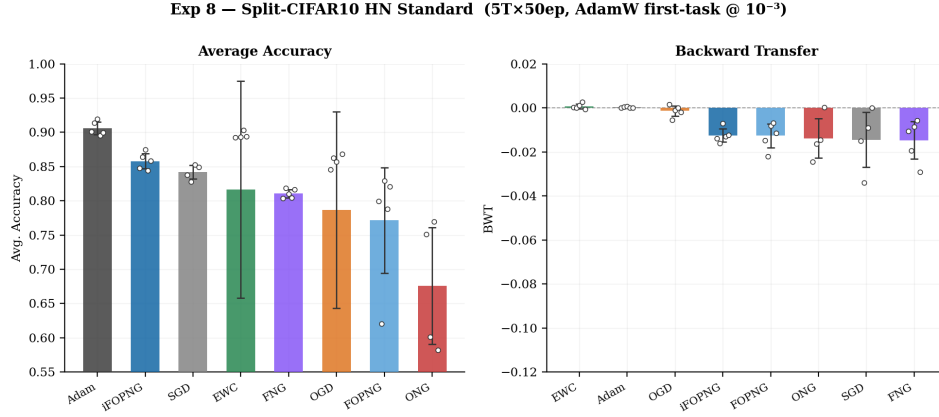


Figure 1: **Split-CIFAR10 hypernetwork (5 tasks, 50 epochs per task, AdamW first-task initialisation at 10^{-3})**. Left: average accuracy (bars show mean over seeds; circles show individual seeds). Right: backward transfer. Methods are sorted by average accuracy on the left panel. Adam defines the projection-free upper frontier; SGD defines the unconstrained lower reference. All projection methods trained with full normalization (Section 5.2) exceed SGD but fall short of Adam.

SUFFOCATED REGIME — SPLIT-MNIST, $d_h = 8$. The embedding dimension $d_h = 8$ was selected via a three-point sweep over $d_h \in \{4, 8, 16\}$, presented as an ablation in Figure 10. That sweep shows the projection-Adam gap is largest at $d_h = 8$ (23.5 pp), falls to 14.0 pp at $d_h = 4$ where seed variance becomes extreme and results near-random, and shrinks to 9.4 pp at $d_h = 16$ as the projection methods partially recover. $d_h = 8$ was retained as the most diagnostically informative level: the failure mode is severe and replicable, but the network has sufficient capacity for the projections to converge consistently.

Figure 2 presents the full method comparison at $d_h = 8$. Adam leads at 93.4%. The Fisher-free projection method OGD is the strongest projection baseline at 86.6%, followed by EWC at 84.5% and SGD at 79.8%. The Fisher-based methods fall sharply below: iFOPNG at 69.9%, FOPNG at 59.9%, and FNG at 54.8% — gaps of 23–39 pp against Adam.

The BWT panel reveals the mechanism underlying this collapse. iFOPNG, FOPNG, and FNG all achieve near-zero backward transfer ($\text{BWT} \approx 0.00$), while Adam has small negative BWT (-0.015) and SGD forgets more substantially (-0.13).

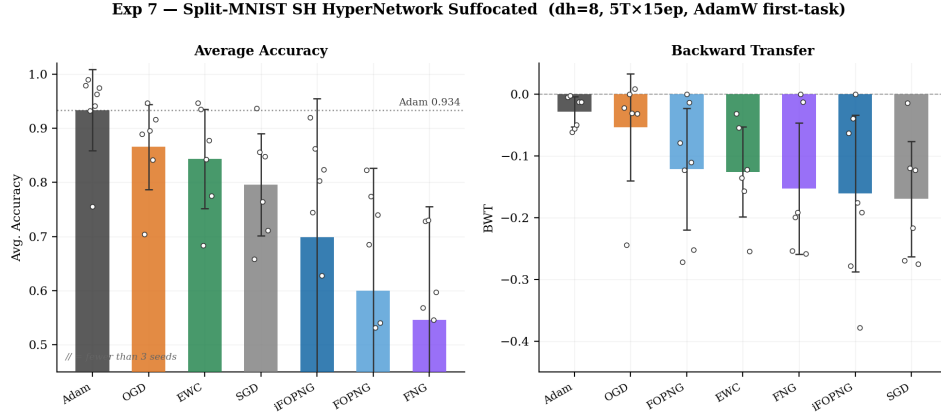


Figure 2: **Split-MNIST SH hypernetwork, suffocated regime ($d_h = 8$, 5 tasks, 15 epochs per task, AdamW first-task initialisation)**. Left: average accuracy (bars show mean over seeds; circles show individual seeds; note that several methods completed fewer than three seeds). Right: backward transfer. The dotted line marks Adam’s mean accuracy (93.4%). Fisher-based projection methods (iFOPNG, FOPNG, FNG) achieve near-zero BWT but much lower accuracy than Adam, revealing a plasticity failure rather than a retention failure in the extreme compression regime. The Fisher-free OGD is the best-performing projection method at 86.6%.

EXTENDED TASK COUNT — SPLIT-CIFAR100. At ten tasks on Split-CIFAR100 (Appendix Figure 14), EWC unexpectedly leads at 45.5%, with Adam second at 42.0% and iFOPNG third at 37.8% (random baseline: 10.0%). This partial rank reversal, in which soft regularization overtakes both Adam and gradient projection, differs from the five-task CIFAR10 pattern and is addressed in the Discussion.

5.2 Sub-RQ2: Normalization Resolves the Gradient Pathology

This subsection reports the three-condition normalization ablation (Split-CIFAR10 hypernetwork; 197 chunks, ≈ 6000 output neurons; unnormalized conditions ran for 10 epochs per task with early stopping, full normalization for 50 epochs). Figure 3 traces the $\log_{10}$ condition number of the projection kernel $G^\top \hat{F} G$ over training steps; Figure 4 shows the resulting average accuracy and BWT under each condition.

WITHOUT NORMALIZATION. The right panel of Figure 3 shows the condition number reaching $10^{10.8}$ at the T₂→T₃ task transition for the most-affected seed, stabilizing near $10^{8.8}$ – $10^{9.0}$ for two others. At this scale, numerical inversion of the projection kernel is unreliable, effectively zeroing the projected gradient so that the method operates as an unconstrained optimizer with unstable Fisher scaling. The result is 72.2% average accuracy

and $\text{BWT} = -0.070$ — worse than plain SGD (84.1%) despite identical hyperparameters. The pathology therefore not only eliminates the benefit of projection but actively degrades performance below the projection-free baseline.

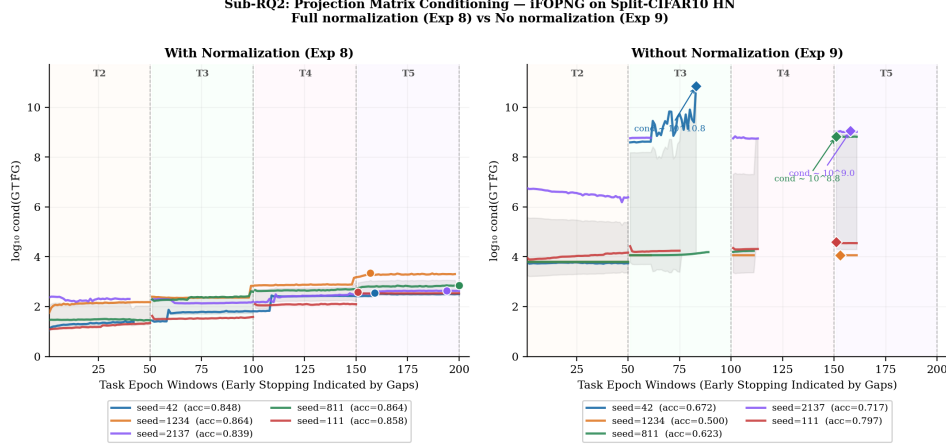


Figure 3: Projection kernel condition number over training — iFOPNG on Split-CIFAR10 HN. Each line is one seed (legend shows final average accuracy). Task boundaries are marked by vertical dashed lines; gaps indicate early stopping. Left (full normalization): $\log_{10} \kappa(G^\top \hat{F} G)$ remains bounded within $[10^{1.3}, 10^{3.3}]$ throughout. Right (no normalization): condition number spikes to $10^{10.8}$ at the $T2 \rightarrow T3$ transition, causing projection failure. Diamond markers indicate end-of-task condition number values.

GRADIENT-ONLY NORMALIZATION. Applying unit- ℓ_2 column normalization to G alone, without rescaling $\hat{F}$, fails to resolve the pathology. Average accuracy falls further to 68.9% ($\text{BWT} = -0.060$) and variance across seeds remains high (Figure 4). As the left panel of Figure 3 illustrates, the raw Fisher magnitudes from the chunked backward passes still span several orders of magnitude, preserving the ill-conditioning of $G^\top \hat{F} G$ even when G itself is normalized. Partial normalization is therefore insufficient.

FULL NORMALIZATION. Combining column-wise ℓ_2 normalization of G with absolute-maximum scaling of $\hat{F}$ fully resolves the pathology. The left panel of Figure 3 shows $\log_{10} \kappa(G^\top \hat{F} G)$ bounded within $[10^{1.3}, 10^{3.3}]$ across all seeds and task transitions — a reduction of more than seven orders of magnitude. Figure 4 shows the result: average accuracy recovers to 85.8% and BWT improves to -0.013 , a 5.7 pp reduction in forgetting relative to no normalization. The improvement is consistent across all three

seeds. Full normalization is adopted for all hypernetwork experiments in this thesis.

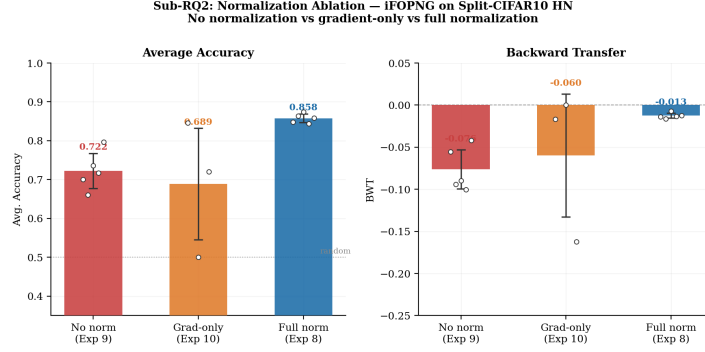


Figure 4: **Normalization ablation — iFOPNG on Split-CIFAR10 HN (Sub-RQ2).** Three conditions: no normalization, gradient-only normalization, and full normalization. Left: average accuracy with mean annotated on each bar. Right: backward transfer. Circles show individual seeds; bars show the mean. The dotted line on the left panel marks random-chance performance (50% for binary per-task classification).

DIAGNOSTIC: SGD FIRST-TASK CONVERGENCE THRESHOLD. A diagnostic experiment (Appendix Figure 15) identifies an inconsistency in the FOPNG paper’s stated first-task protocol. At FOPNG’s learning rate of $\eta = 10^{-5}$ on Split-MNIST, five epochs of SGD training yield only $\approx 21\%$ accuracy — barely above the 50% random baseline. The empirical convergence threshold lies between 5×10^{-5} and 10^{-4} . All standalone experiments in this thesis use $\eta = 10^{-3}$ for first-task SGD, deviating from but empirically improving upon the original protocol.

5.3 Sub-RQ3: Parameter Inertia via the Combined Fisher Metric

This subsection evaluates whether replacing FOPNG’s task-specific natural gradient metric $\hat{F}_{\text{new}}$ with the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ systematically improves task retention across benchmarks. Figure 5 presents mean average accuracy for both methods across all six benchmarks.

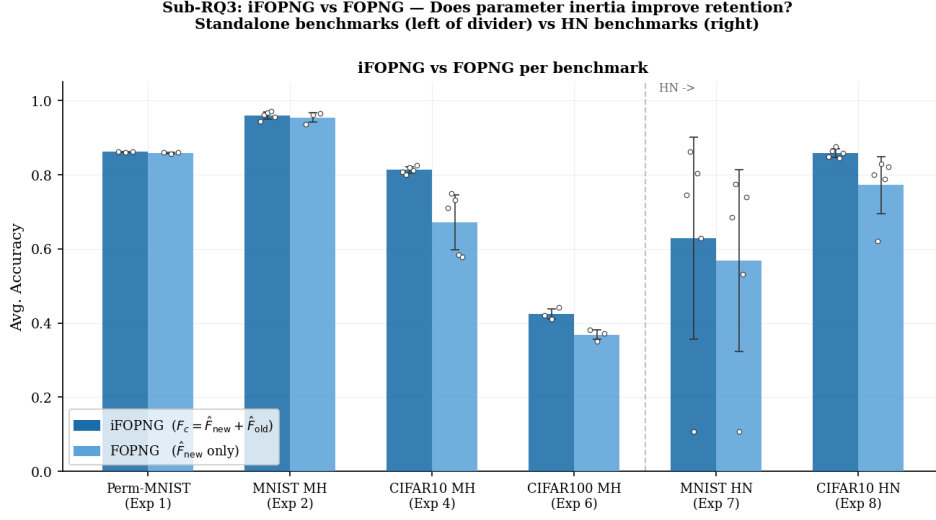


Figure 5: **iFOPNG vs. FOPNG across six benchmarks (Sub-RQ3).** Grouped bars show mean average accuracy; circles show individual seeds; error bars show ± 1 standard deviation. The dashed vertical line separates standalone (left) from hypernetwork (right) settings. iFOPNG uses the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$; FOPNG uses only $\hat{F}_{\text{new}}$.

On the two simplest benchmarks — Permuted-MNIST and Split-MNIST multi-head — the two methods are nearly indistinguishable (86.0% vs. 85.9% and 95.6% vs. 95.2% respectively).

Both tasks are low-dimensional with limited inter-task Fisher overlap.

A clear separation emerges on harder benchmarks. On Split-CIFAR10 multi-head, iFOPNG reaches 81.1% versus FOPNG’s 67.4%, a gap of 14.1 pp. On Split-CIFAR100 multi-head the advantage is 5.6 pp (42.7% vs. 37.1%). Detailed per-task trajectories for these standalone references are provided in Appendix B (Figures 8 and 9).

The advantage carries over into the hypernetwork settings: 9.8 pp on Split-MNIST HN (70.0% vs. 60.0%) and 8.6 pp on Split-CIFAR10 HN (85.6% vs. 77.5%). Across all six benchmarks, the retention advantage of iFOPNG scales robustly with task complexity and cumulative Fisher overlap.

5.4 Sub-RQ4: Fisher Accumulation Operator

This subsection compares two strategies for accumulating F_{old} across tasks under the iFOPNG combined Fisher metric: the element-wise maximum operator (MAX) and an exponential moving average (EMA). Both are evaluated on a twenty-task Permuted-MNIST sequence with five random seeds. Figure 6 presents the average accuracy and backward transfer

trajectories over all twenty tasks; the per-seed statistical comparison is provided in Appendix Figure 16.

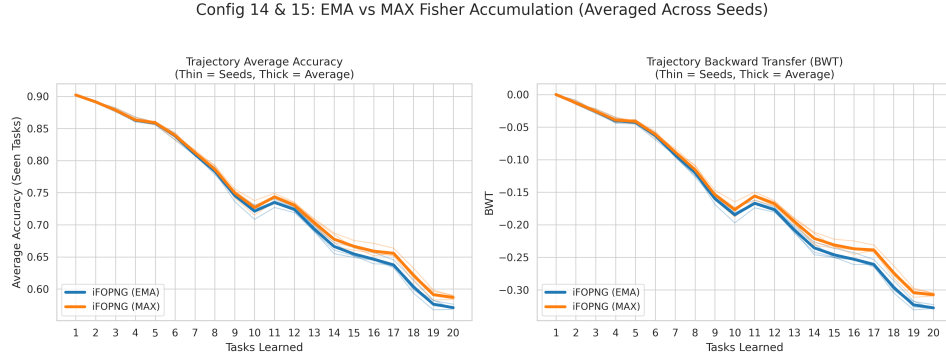


Figure 6: **EMA vs. MAX Fisher accumulation trajectories on 20-task Permuted-MNIST (Sub-RQ4), $n = 5$ seeds.** Thick lines show the seed-averaged trajectory; thin lines show individual seeds. Left: average accuracy over tasks seen so far, evaluated after each task is trained. Right: backward transfer over the same sequence. MAX leads EMA by a consistent margin across all five seeds; both strategies produce closely tracking trajectories throughout the full twenty-task sequence.

A paired t -test on final average accuracy establishes MAX as the superior operator: $t(4) = 14.43$, $p < 0.001$, Cohen’s $d = 7.22$, with MAX leading by a consistent 1.51–1.65 pp across every seed pair without exception (Appendix Figure 16). The result is statistically unambiguous and replicates across all five seeds. In absolute terms the margin is practically negligible — 1.60 pp on a benchmark where both variants decline by more than 30 pp over the full sequence — but the direction is consistent and the effect size large, leaving no ambiguity about which operator to prefer.

The MAX operator is therefore preferred as the simpler, marginally better, and hyperparameter-free alternative.

6 DISCUSSION

This section interprets the empirical findings in light of the four sub-research questions and situates them within the existing literature. The overarching goal was to determine whether gradient projection methods can be fruitfully integrated into chunked hypernetworks for continual learning, and to identify the structural conditions under which this integration succeeds or fails.

6.1 Sub-RQ1: Constructive Synergy or Architectural Conflict?

Projection methods do not outperform a well-tuned regularized hypernetwork optimized via Adam. The gap of 4.5 pp at task 5 in favor of Adam is consistent across seeds and widens substantially in the suffocated regime ($d_h = 8$, Split-MNIST SH), where it reaches 20–40 pp. Critically, the BWT data reveal that in the suffocated regime this is a *plasticity* failure, not a *retention* failure: Fisher-based projection methods achieve near-zero backward transfer alongside collapsed accuracy, meaning the curvature-weighted constraint exhausts the available gradient budget for new task learning rather than permitting forgetting of prior ones. OGD’s relative resilience at $d_h = 8$ is consistent with this reading: without Fisher weighting its subspace constraint is less aggressive, leaving more gradient capacity for forward transfer. We attribute the broader gap to a fundamental architectural incompatibility. Projection methods operate by modifying the raw gradient before the optimizer state update, which means the first-order momentum estimate and the adaptive scaling factor maintained by Adam are effectively discarded at each projected step. In the highly nonlinear, compressed weight space of a hypernetwork meta-generator, momentum is not merely a convergence accelerant — it is a structural regularizer that prevents the optimizer from committing prematurely to poorly-conditioned gradient directions. Re-initializing the optimizer’s internal state at each step destabilizes the generative manifold and reduces the quality of the generated weights, particularly under extreme bottleneck compression. This finding is consistent with [Hu, Yu, Cheng, Liu, and Song \(2026\)](#), who demonstrate that gradient modification methods interact adversely with Adam’s internal state under certain curvature regimes, and aligns with the broader observation that projection-based continual learning methods were originally designed for SGD-like updates.

The ten-task Split-CIFAR100 result introduces a partial rank reversal that complicates the Sub-RQ1 conclusion: EWC unexpectedly leads at 45.5%, overtaking both Adam (42.0%) and iFOPNG (37.8%). This inversion suggests that at higher task counts, soft functional regularization in output space may dominate hard gradient projection because the projection memory cannot hold sufficient historical directions within a fixed column budget. As the number of tasks grows, the gradient subspace spanned by G becomes an increasingly coarse approximation to the full historical interference manifold, while the functional regularizer in output space grows more informative with each additional stored weight snapshot. Whether this effect persists at the twenty-task scale, and whether an expanded memory budget would recover the projection advantage, remains to be investigated.

FIRST-TASK INITIALIZATION AND GRADIENT CONTAMINATION. The comparison between AdamW and Adam first-task initialization confirmed the contamination hypothesis for gradient projection methods in the hyper-network setting. Standard Adam couples weight decay into the gradient estimate as an additive ℓ_2 term, so both the gradient memory G and the Fisher diagonal $\hat{F}$ collected at the end of task 1 inherit a component aligned with the weight magnitude vector rather than the task loss curvature. This contaminates the projection subspace: downstream tasks are projected away from directions spuriously marked as historical but actually reflecting optimizer regularization rather than task geometry. AdamW initialization, which applies weight decay as a decoupled parameter-shrinkage step, consistently produced lower first-task trajectory variance and a cleaner initial projection basis. This finding carries a practically important implication for any system bootstrapping a projection memory from first-task gradients: the decoupled optimizer should be preferred at that stage regardless of which optimizer is used in subsequent tasks.

6.2 Sub-RQ2: Projection-Scoped Normalization Resolves the Pathology

This result contributes a practical engineering solution that is necessary for any future work combining hard projection constraints with chunked meta-learners, regardless of which projection algebra is employed. The normalization is applied locally within the projection computation and does not alter the underlying optimization landscape: subspace membership is invariant to positive scalar rescaling of basis vectors, so neither operation changes the geometry of the projection. The fundamental problem is therefore purely numerical rather than conceptual: the intersection of chunked hypernetworks and gradient projection is geometrically coherent but arithmetically fragile without explicit local rescaling.

6.3 Sub-RQ3: Parameter Inertia via the Elastic Fisher Preconditioner

The iFOPNG advantage scales monotonically with benchmark difficulty across all six settings (Figure 5): negligible where inter-task Fisher overlap is low, substantial on high-overlap benchmarks. F_c inflates the effective curvature of historically important coordinates, suppressing over-aggressive natural gradient steps without any penalty term or stored reference point.

This mechanism stands in direct contrast to EWC, which achieves a similar dampening effect via an explicit quadratic penalty pulling parameters toward a reference point. iFOPNG applies no such restoring force and maintains no reference: a parameter that mattered before is not pulled back, it is made harder to displace. The empirical advantage of iFOPNG

over FOPNG on high-overlap benchmarks confirms that the rigidity of the pure projection boundary is a genuine limitation of FOPNG, and that lifting it via the combined metric F_c is a principled and effective remedy. Nevertheless, iFOPNG does not close the gap to Adam, confirming that the elastic preconditioner addresses rigidity but cannot recover the momentum mechanism that Adam leverages for efficient navigation of the generative manifold.

6.4 Sub-RQ4: Fisher Accumulation Operator

This null result is itself informative. The theoretical argument for EMA was that monotonic MAX growth would impose a steadily increasing effective mass on parameters that were important to any prior task, eventually over-constraining the natural gradient and harming plasticity on new tasks. The absence of this effect on Permuted-MNIST suggests that, under the normalization scheme applied here (which rescales both Fisher terms by the current maximum of F_c), the absolute scale of F_{old} does not enter the projection computation — only its relative distribution does. The max-normalized Fisher is therefore insensitive to whether the accumulation operator is decaying or non-decaying, as long as the distribution of importance across parameters is preserved.

Two limitations of this finding should be noted. First, Permuted-MNIST is a structurally homogeneous benchmark in which all tasks share the same label space and architecture; inter-task Fisher overlap is low and approximately uniform across tasks. On benchmarks with heterogeneous task distributions and non-uniform overlap, the two operators may diverge as the historical Fisher accumulates disproportionate mass in task-specific parameter subspaces. Second, the experiment used a fixed EMA decay of $\alpha = 0.5$; an adaptive schedule that increases decay as the task sequence grows longer may produce a more favorable comparison for EMA under long-horizon retention.

6.5 Limitations

Several limitations should be acknowledged.

First, the hypernetwork results presented here are restricted to Split-MNIST and Split-CIFAR10 with five tasks. Longer task sequences may produce different relative orderings: in particular, the advantage of hard projection constraints over soft functional regularization may grow with sequence length, as soft penalties are known to suffer from compounding output drift (Farquhar & Gal, 2018), while the projection subspace scales with the gradient budget rather than the task count. The preliminary ten-

task Split-CIFAR100 result, in which EWC overtakes both Adam and iFOPNG, is suggestive but insufficient to establish a trend; a systematic sweep over task count is warranted.

Second, the hypernetwork architecture used in this study is intentionally shallow, with small hidden dimensions ($d_h \in \{8, 16\}$) and embedding dimensions of 32. Larger hypernetworks, such as those used for language model weight generation (Chauhan, Zhou, Lu, Molaei, & Clifton, 2024), may exhibit qualitatively different gradient accumulation dynamics — both in terms of the severity of the normalization pathology and in the relative contribution of momentum to meta-generator stability. The suffocated regime ($d_h = 8$) explored here provides an empirical lower bound on expressivity, but an upper bound in a high-capacity regime remains uncharted.

Third, the proposed normalization scheme — QR orthonormalization of the gradient basis combined with absolute-maximum scaling of the metric tensor F_c — is heuristic in nature. It discards the eigenvalue structure of the pullback metric and applies isotropic rather than curvature-aware rescaling to the chunk embedding directions. The principled alternative is the Riemannian gradient under the pullback metric $G_i = J_i^\top J_i$, where $J_i = \partial W_i / \partial c_i$ is the Jacobian of the chunk-generating map:

$$\Delta c_i = -\eta (J_i^\top J_i + \lambda I)^{-1} \nabla_{c_i} \mathcal{L}. \quad (19)$$

This preconditioner was implemented using batched forward-mode automatic differentiation across all K chunks simultaneously. When applied to the raw gradient *prior* to the FOPNG projection step, however, it consistently degraded continual learning performance in preliminary evaluation: backward transfer worsened by approximately 0.33 units on Split-MNIST SH relative to the heuristic baseline. We attribute this to a metric mismatch: the FOPNG projection operator is calibrated for gradients in Euclidean parameter space, and preconditioning only the chunk embedding component before projection produces a geometrically inconsistent input to the kernel $G^\top \hat{F} G$. The principled resolution — applying the natural gradient to the projected update v^* rather than the raw gradient, so that the projection sees Euclidean inputs and the Riemannian step follows — is left to future work.

7 CONCLUSION

This thesis investigated the extent to which gradient projection methods can be successfully integrated into chunked hypernetwork architectures for continual learning, and identified the structural conditions under which that integration yields benefit or harm. The primary research question

decomposed into four sub-questions addressing constructive synergy, gradient normalization, the elastic Fisher metric, and the Fisher accumulation operator.

SUB-RQ1 — PARTIAL SYNERGY BOUNDED BY THE MOMENTUM PROBLEM. The integration is constructive but incomplete. Every normalized projection variant surpasses unconstrained SGD in the hypernetwork setting, confirming that hard geometric constraints add genuine retention value. No projection variant matches Adam, however, and the gap widens dramatically under extreme compression — the loss of adaptive momentum is the binding constraint, not the projection geometry itself. Resolving this will require projection-compatible adaptive optimizers rather than post-hoc gradient modification (Hu et al., 2026).

SUB-RQ2 — NORMALIZATION IS A NECESSARY CONDITION. Without explicit local rescaling, projection methods are numerically incompatible with chunked hypernetworks: matrix inversion in the projection kernel collapses completely, degrading performance below plain SGD. QR orthonormalization of the gradient basis combined with absolute-maximum Fisher scaling resolves the pathology entirely. This normalization scheme is a necessary engineering precondition for any future combination of gradient projection with chunked meta-learners, regardless of which projection algebra is employed.

SUB-RQ3 — PARAMETER INERTIA IMPROVES RETENTION ON HIGH-OVERLAP BENCHMARKS. Replacing FOPNG’s task-specific Fisher metric with the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ consistently improves task retention when inter-task Fisher overlap is substantial. The advantage of iFOPNG over FOPNG grows monotonically with benchmark difficulty: negligible on Permuted-MNIST (+0.1 pp), substantial on Split-CIFAR10 multi-head (+14.1 pp), and meaningful in the hypernetwork settings (+9.8 pp and +8.6 pp). iFOPNG thereby provides a principled and practically effective improvement over FOPNG at no additional hyperparameter cost.

SUB-RQ4 — MAX ACCUMULATION IS THE CLEAR PRACTICAL CHOICE. On a twenty-task Permuted-MNIST sequence evaluated across five seeds, the element-wise maximum (MAX) and exponential moving average (EMA) accumulation strategies produced nearly parallel accuracy and backward transfer trajectories throughout the full sequence. A paired t -test confirms that MAX is the superior operator: $t(4) = 14.43$, $p < 0.001$, Cohen’s $d = 7.22$, with MAX leading by a consistent 1.51–1.65 pp across every seed pair. The effect is systematic and statistically unambiguous, though small

in absolute terms: 1.60 pp on a benchmark where both strategies decline by more than 30 pp over the full sequence — practically negligible in deployment terms. The MAX operator is therefore preferred as the simpler, marginally better, and hyperparameter-free alternative.

IMPLICATIONS FOR THE FIELD. The results establish a clear empirical boundary for the current generation of gradient projection methods: they can be made numerically stable in chunked hypernetwork settings via local rescaling, and their Fisher geometry can be improved via elastic accumulation, but the momentum incompatibility prevents them from reaching the performance frontier of adaptive unconstrained optimizers in generative parameter spaces. The normalization contribution is directly applicable to any method that maintains a gradient subspace memory alongside a chunked weight generator. The iFOPNG contribution extends naturally to any continual learning setting where Fisher overlap accumulates across tasks. Together, they narrow the gap between projection-based and regularization-based approaches, and provide a principled foundation for the next step: a projected adaptive optimizer that respects both the geometric constraints of the projection and the curvature structure that adaptive moment estimates capture.

FUTURE DIRECTIONS. The most direct extension is a projected variant of Adam that maintains momentum across projected steps rather than discarding it at each update, addressing the primary mechanism identified in Sub-RQ1; this direction has been partially explored in concurrent work (Hu et al., 2026).

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REFERENCES

- Amari, S.-i. (1998). Natural gradient works efficiently in learning. , 10(2), 251–276. Retrieved 2026-05-31, from <https://ieeexplore.ieee.org/document/6790500> doi: 10.1162/089976698300017746
- Bennani, M. A., Doan, T., & Sugiyama, M. (2020). *Generalisation guarantees for continual learning with orthogonal gradient descent* (No. arXiv:2006.11942). arXiv. Retrieved 2026-04-22, from <http://arxiv.org/abs/2006.11942> doi: 10.48550/arXiv.2006.11942
- Chauhan, V. K., Zhou, J., Lu, P., Molaei, S., & Clifton, D. A. (2024). A brief review of hypernetworks in deep learning. , 57(9), 250. Retrieved 2026-02-21, from <http://arxiv.org/abs/2306.06955> doi: 10.1007/s10462-024-10862-8
- De Lange, M., Aljundi, R., Masana, M., Parisot, S., Jia, X., Leonardis, A., ... Tuytelaars, T. (2022). A continual learning survey: Defying forgetting in classification tasks. , 44(7), 3366–3385. Retrieved 2026-05-16, from <https://ieeexplore.ieee.org/document/9349197/> doi: 10.1109/TPAMI.2021.3057446
- Farajtabar, M., Azizan, N., Mott, A., & Li, A. (2020). *Orthogonal gradient descent for continual learning* (No. arXiv:1910.07104). arXiv. Retrieved 2026-02-23, from <http://arxiv.org/abs/1910.07104> doi: 10.48550/arXiv.1910.07104
- Farquhar, S., & Gal, Y. (2018). *Towards robust evaluations of continual learning* (No. arXiv:1805.09733). arXiv. Retrieved 2026-04-05, from <http://arxiv.org/abs/1805.09733> doi: 10.48550/arXiv.1805.09733
- Fuente, N. D. L., Pilligua, M., Vidal, D., Soutiff, A., Curreli, C., Cremers, D., & Barsky, A. (2025). *Prototype augmented hypernetworks for continual learning* (No. arXiv:2505.07450). arXiv. Retrieved 2026-02-28, from <http://arxiv.org/abs/2505.07450> doi: 10.48550/arXiv.2505.07450
- Garg, I., Kolhe, N., Peng, A., & Gopalam, R. (2026). *Fisher-orthogonal projected natural gradient descent for continual learning* (No. arXiv:2601.12816). arXiv. Retrieved 2026-02-21, from <http://arxiv.org/abs/2601.12816> doi: 10.48550/arXiv.2601.12816
- Grossberg, S. (1980). How does a brain build a cognitive code? , 87(1), 1–51. (Place: US Publisher: American Psychological Association) doi: 10.1037/0033-295X.87.1.1
- Ha, D., Dai, A., & Le, Q. V. (2016). *HyperNetworks* (No. arXiv:1609.09106). arXiv. Retrieved 2026-02-21, from <http://arxiv.org/abs/1609.09106> doi: 10.48550/arXiv.1609.09106
- Hemati, H., Lomonaco, V., Bacciu, D., & Borth, D. (2023). *Partial hypernetworks for continual learning* (No. arXiv:2306.10724). arXiv. Re-

- trieved 2026-02-28, from <http://arxiv.org/abs/2306.10724> doi: 10.48550/arXiv.2306.10724
- Hu, Y., Yu, Z., Cheng, Z., Liu, W., & Song, L. (2026). *Hidden failure modes of gradient modification under adam in continual learning, and adaptive decoupled moment routing as a repair* (No. arXiv:2604.22407). arXiv. Retrieved 2026-05-16, from <http://arxiv.org/abs/2604.22407> doi: 10.48550/arXiv.2604.22407
- Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., ... Hadsell, R. (2017). Overcoming catastrophic forgetting in neural networks. , 114(13), 3521–3526. Retrieved 2026-02-23, from <http://arxiv.org/abs/1612.00796> doi: 10.1073/pnas.1611835114
- McCloskey, M., & Cohen, N. J. (1989). *Catastrophic interference in connectionist networks: The sequential learning problem* (Vol. 24). Academic Press. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0079742108605368> doi: [https://doi.org/10.1016/S0079-7421\(08\)60536-8](https://doi.org/10.1016/S0079-7421(08)60536-8)
- Oswald, J. v., Henning, C., Grewe, B. F., & Sacramento, J. (2020). *Continual learning with hypernetworks* (No. arXiv:1906.00695). arXiv. Retrieved 2026-02-21, from <http://arxiv.org/abs/1906.00695> doi: 10.48550/arXiv.1906.00695
- Yadav, Y., Mendoza, P., & Korrapati, J. (2025). *ONG: Orthogonal natural gradient descent* (No. arXiv:2508.17169). arXiv. Retrieved 2026-06-04, from <http://arxiv.org/abs/2508.17169> doi: 10.48550/arXiv.2508.17169

8 APPENDIX

APPENDIX A: IFOPNG THEOREM AND PROOF

Notation

Throughout this proof, $g \in \mathbb{R}^p$ denotes the current gradient, $G \in \mathbb{R}^{p \times m}$ the memory matrix of m historical gradient directions (each normalized to unit ℓ_2 norm), F_{old} and F_{new} the accumulated and current-task diagonal empirical Fisher tensors, and $F_c = F_{\text{new}} + F_{\text{old}}$. We treat diagonal Fisher matrices as vectors acting by element-wise multiplication; all matrix operations below are defined accordingly.

Assumption 1 (Regularity). F_c is positive definite and the matrix $A = G^\top F_{\text{old}} F_c^{-1} F_{\text{old}} G$ is invertible.

Theorem 1 (iFOPNG Update). *Under Assumption 1, the solution to*

$$\begin{aligned} \min_u \quad & \|g - F_c^{-1}u\|_{F_c}^2 \\ \text{s.t.} \quad & F_{\text{old}}^{-1}u \in \text{Col}(G), \\ & \|v\|_{F_c} = \eta, \end{aligned} \tag{20}$$

with parameter update $v = c F_c^{-1}(g - u)$, is given by (16), where $P = I - F_{\text{old}} G A^{-1} G^\top F_{\text{old}}$.

Proof. **Step 1.** The constraint $F_{\text{old}}^{-1}u \in \text{Col}(G)$ implies $u = F_{\text{old}} G \alpha$ for some $\alpha \in \mathbb{R}^m$.

Step 2. Substituting and expanding using $\|w\|_{F_c}^2 = w^\top F_c w$:

$$\|g - F_c^{-1}u\|_{F_c}^2 = g^\top F_c g - 2g^\top F_{\text{old}} G \alpha + \alpha^\top A \alpha. \tag{21}$$

Step 3. Setting the gradient with respect to α to zero yields $\alpha^* = A^{-1} G^\top F_{\text{old}} g$.

Step 4. Substituting back: $u^* = F_{\text{old}} G A^{-1} G^\top F_{\text{old}} g$, so $g - u^* = Pg$.

Step 5. Applying the trust-region constraint $\|v\|_{F_c} = \eta$ to $v = c F_c^{-1}Pg$ gives $c = \eta / \sqrt{(Pg)^\top F_c^{-1}Pg}$, yielding the stated update. $\square$

APPENDIX B: STANDALONE TARGET-NETWORK RESULTS

This appendix presents the full method comparison on standalone target networks — configurations where a conventional MLP is trained directly via gradient projection, without any hypernetwork wrapper or functional

regularizer. These results establish the baseline behaviour of each projection method in its intended setting, prior to the hypernetwork integration examined in the main text. Four benchmark streams are reported: Permuted-MNIST (5 tasks) and Split-MNIST multi-head (5 tasks) in Figure 7, Split-CIFAR10 multi-head (5 tasks) in Figure 8, and Split-CIFAR100 multi-head (10 tasks) in Figure 9. Bars show mean accuracy over seeds; circles show individual seeds.

Standalone Target-Network Benchmarks — MNIST

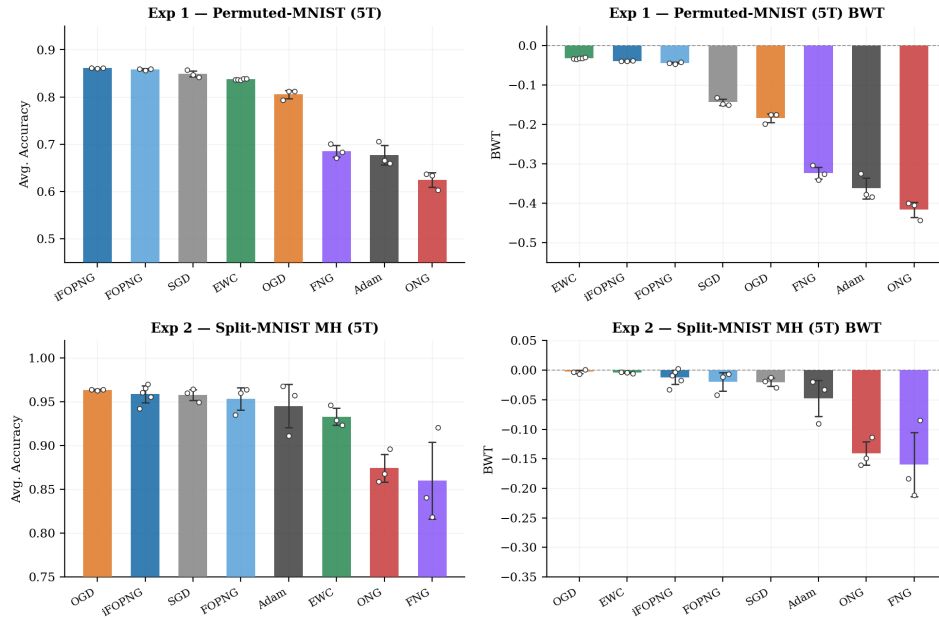


Figure 7: **Standalone target-network results — Permuted-MNIST (5 tasks, top row) and Split-MNIST multi-head (5 tasks, bottom row).** Left column: average accuracy. Right column: backward transfer. Bars show mean over seeds; circles show individual seeds. *Permuted-MNIST*. iFOPNG and FOPNG lead at 86.1% and 85.8% respectively with near-zero backward transfer, closely followed by SGD (84.9%) and EWC (83.8%). Adam (68.2%) and ONG (62.6%) exhibit substantially negative BWT (≈ -0.36 and ≈ -0.45), reflecting severe catastrophic forgetting without a projection or regularization mechanism. *Split-MNIST MH*. Most methods cluster near the ceiling ($\geq 95\%$), reflecting the low difficulty of the binary per-task splits. OGD, iFOPNG, SGD, and FOPNG lead (96.1–95.6%); ONG and FNG are the notable exceptions (87% and 86%) with strongly negative BWT, indicating convergence instability on this stream.

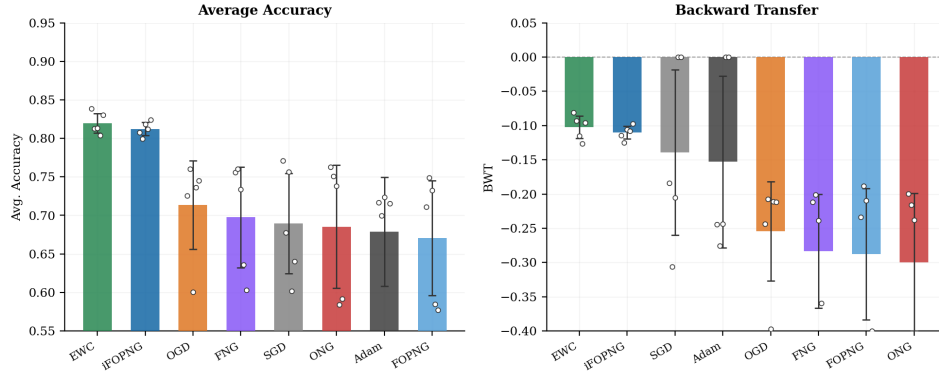
Exp 4 — Split-CIFAR10 MH Standalone (5T×5ep, Adam first-task @ 10^{-3})

Figure 8: **Standalone target-network results — Split-CIFAR₁₀ multi-head (5 tasks, 5 epochs per task, Adam first-task at 10^{-3}).** Left: average accuracy. Right: backward transfer. EWC leads at 81.9% and iFOPNG is a close second at 81.1%; both exhibit moderate forgetting ($BWT \approx -0.09$). The remaining methods fall to 67–71% with substantially higher seed variance and stronger forgetting. FOPNG is the weakest projection method at 67.4%, a 13.7 pp gap below iFOPNG that motivates the Sub-RQ₃ comparison. Adam (68.1%) performs comparably to the weaker projection methods, confirming that the unconstrained baseline is not dominant on harder standalone benchmarks.

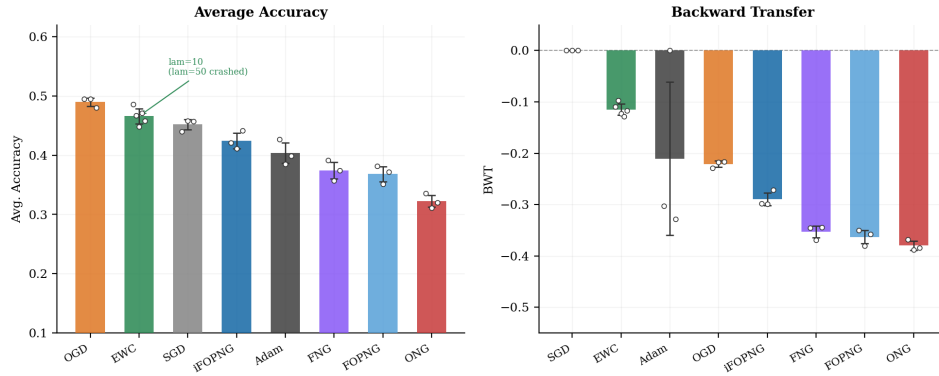
Exp 6 — Split-CIFAR100 MH Standalone (10T×10ep, SGD first-task @ 10^{-2})

Figure 9: **Standalone target-network results — Split-CIFAR₁₀₀ multi-head (10 tasks, 10 epochs per task, SGD first-task at 10^{-2}).** Left: average accuracy. Right: backward transfer. EWC was run with $\lambda = 10$; the $\lambda = 50$ configuration produced collapsed runs and is excluded. OGD leads at 49.2%, followed by EWC ($\lambda=10$, 46.8%) and SGD (45.4%). iFOPNG reaches 42.7%, a 5.6 pp advantage over FOPNG (37.1%), consistent with the Sub-RQ₃ finding that parameter inertia provides increasing benefit as task count and Fisher overlap grow. ONG has the lowest average accuracy (32.3%) and the most negative BWT (≈ -0.41), indicating near-complete forgetting across the ten-task sequence. Random-chance performance is 10% (10-way classification per task).

APPENDIX C: HYPERNETWORK ARCHITECTURE ABLATIONS

This appendix documents three architectural ablations that motivated the hypernetwork configurations used in the main experiments. Figure 10 establishes the diagnostic capacity level $d_h = 8$ for the suffocated Split-MNIST regime (Sub-RQ₁, Benchmark 1). Figure 11 presents the joint sweep over regularization strength β and chunk size c used to select the operating point for all main hypernetwork experiments. Figure 12 traces the per-task evolution of average accuracy and backward transfer for the Split-CIFAR10 standard hypernetwork across all five task transitions.

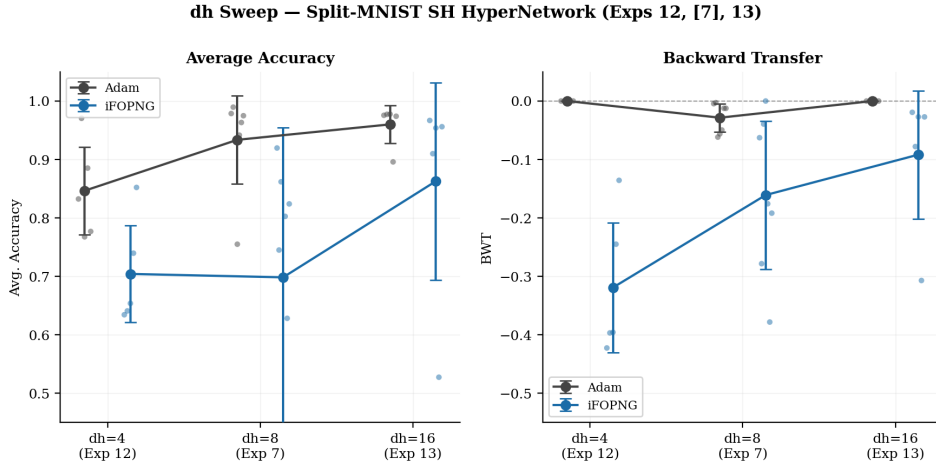


Figure 10: **Hidden-dimension sweep — Split-MNIST SH hypernetwork** ($d_h \in \{4, 8, 16\}$, 5 tasks, 15 epochs per task, AdamW first-task initialisation at 10^{-3}). Left: average accuracy; right: backward transfer. Thick markers show seed means; error bars show ± 1 standard deviation; individual seeds are shown as dots. Only Adam and iFOPNG are plotted to isolate the projection-baseline gap across capacity levels. Three capacity levels are compared. At $d_h = 4$ (Exp 12), the Adam-iFOPNG gap is 14.0 pp, but individual seed variance for iFOPNG is extreme (range ≈ 0.25), indicating that the bottleneck is too severe for reliable projection convergence. At $d_h = 8$ (Exp 7), the gap widens to 23.5 pp and variance contracts substantially: seeds cluster in the range 0.63–0.92 for Adam and 0.63–0.75 for iFOPNG, making the failure mode severe and consistently replicable. At $d_h = 16$ (Exp 13), projection methods partially recover — the gap narrows to 9.4 pp — as the larger gradient budget reduces the relative cost of subspace projection. The backward transfer panel mirrors this pattern: iFOPNG BWT improves monotonically with capacity (≈ -0.32 at $d_h = 4$, ≈ -0.17 at $d_h = 8$, ≈ -0.11 at $d_h = 16$), while Adam remains near zero throughout. $d_h = 8$ was retained as the primary diagnostic level: the plasticity failure is both severe and consistently replicable, whereas $d_h = 4$ yields near-random results in some seeds and $d_h = 16$ partially obscures the failure mode.

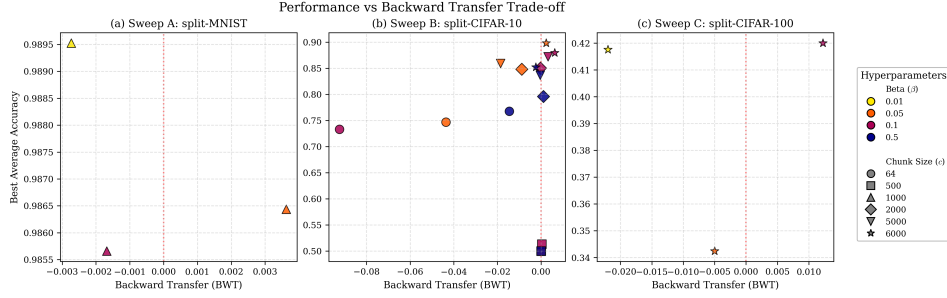


Figure 11: Beta and chunk-size sweep — hypernetwork performance vs. backward transfer trade-off across three benchmarks (Exps 16–18). Each panel plots mean average accuracy (y -axis) against backward transfer (x -axis) for all evaluated combinations of regularization strength $\beta \in \{0.01, 0.05, 0.1, 0.5\}$ and chunk size $c \in \{64, 500, 1000, 2000, 5000, 6000\}$. Marker shape encodes chunk size; marker colour encodes β . The vertical red dashed line marks $\text{BWT} = 0$; points to its left exhibit net retention, points to its right exhibit net forgetting. The probe method is Adam with the functional regularizer active. Split-MNIST (panel a) is insensitive to both hyperparameters: all combinations achieve 98.55–98.95% accuracy with BWT spanning only ≈ 0.006 , indicating that the task is too simple to discriminate architectures. Split-CIFAR₁₀ (panel b) shows clear structure. Small chunk sizes ($c \in \{64, 500\}$) cluster in the lower-left regardless of β , reaching only 74–75% accuracy; larger chunks progressively shift the Pareto frontier upward. The combination $c = 6000$, $\beta = 0.1$ sits on the efficiency frontier: it achieves 87–88% average accuracy with small negative BWT, providing the best retention–plasticity balance. The combination $c = 6000$, $\beta = 0.5$ yields similar accuracy but with marginally higher forgetting. Split-CIFAR₁₀₀ (panel c) confirms the selection: $c = 6000$, $\beta = 0.5$ achieves the highest accuracy ($\approx 42\%$) but at the cost of slightly positive BWT, suggesting the stronger regularizer over-constrains plasticity on a 10-task sequence. $c = 6000$, $\beta = 0.1$ achieves $\approx 34\%$ with negative BWT and was therefore selected as the operating point for all main hypernetwork experiments, providing consistent Pareto-frontier performance across both CIFAR benchmarks without the positive-BWT artefact observed at higher β .

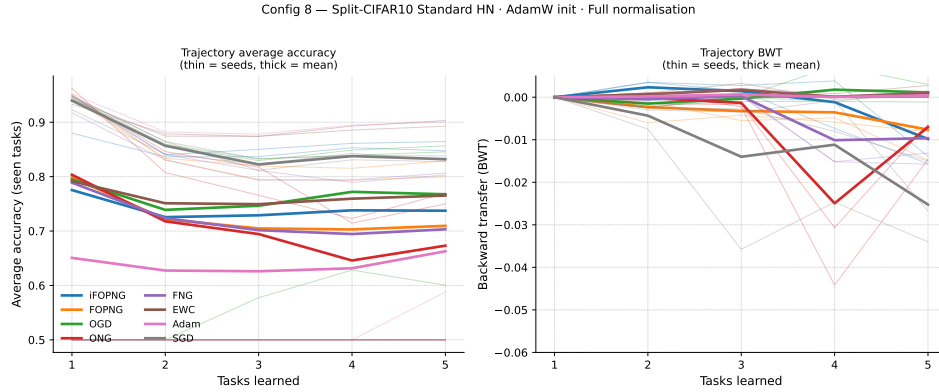


Figure 12: **Per-task learning trajectory — Split-CIFAR10 standard hypernetwork (Config 8, 5 tasks, 50 epochs per task, AdamW first-task initialisation, full normalisation)**. Left: average accuracy over all tasks seen so far, evaluated after each task is trained (x -axis = number of tasks learned). Right: backward transfer over the same sequence. Thick lines show the seed mean; thin lines show individual seeds. After task 1 all methods achieve high single-task accuracy (0.78–0.95), since no forgetting has yet occurred. The stratification that determines the final ordering in the main text (Figure 1) is established by task 2 and remains largely stable thereafter. Adam (pink) maintains the highest trajectory throughout, ending at ≈ 0.84 and separating from the projection cluster at task 2. SGD (grey) tracks Adam closely for tasks 1–3 before diverging slightly downward, reflecting the onset of forgetting without momentum. iFOPNG (blue), EWC (brown), and OGD (green) form a mid-tier cluster between 0.71 and 0.78 from task 2 onward. ONG (red) undergoes a pronounced dip at task 4 (individual seeds visible in the thin lines), partially recovering to ≈ 0.67 by task 5; this instability is consistent with the high seed variance for ONG reported in the main text. The backward transfer panel confirms that the functional regularizer suppresses forgetting across all methods: BWT values remain within $[-0.03, +0.01]$ throughout the full sequence. The accuracy gap between Adam and projection methods therefore reflects a deficit in forward transfer to newly seen tasks rather than differential forgetting of earlier ones — a conclusion consistent with the per-task heatmaps in Appendix D (Figure 13).

APPENDIX D: ADDITIONAL EXPERIMENT FIGURES

This appendix collects four figures that are referenced in the Results section but omitted from the main text for space. Figure 13 provides the per-task accuracy matrix for the Split-CIFAR10 hypernetwork experiment. Figure 14 shows the full method comparison on the ten-task Split-CIFAR100 hypernetwork benchmark. Figure 15 documents the SGD first-task convergence diagnostic. Figure 16 provides the per-seed paired comparison underlying the Sub-RQ4 statistical test.

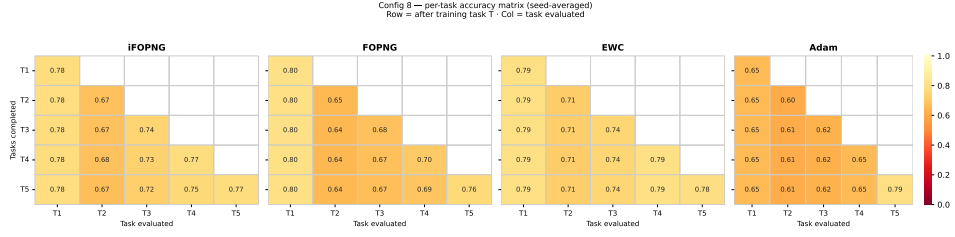


Figure 13: **Per-task accuracy matrix — Split-CIFAR10 hypernetwork, seed-averaged (Sub-RQ1).** Each matrix shows the accuracy on task j (column) after completing training through task i (row). Only the lower triangle is populated; the diagonal gives the just-trained task’s accuracy and off-diagonal entries measure retention of earlier tasks. Four methods are shown: iFOPNG, FOPNG, EWC, and Adam. The T1 column is the primary retention indicator. iFOPNG, FOPNG, and EWC hold T1 accuracy stable at 0.78, 0.80, and 0.79 respectively across all five task transitions. Adam retains T1 at only 0.65 — 0.13 pp lower — indicating that the unconstrained adaptive optimizer forgets more of the first task than any of the projection or regularization methods. However, Adam achieves the highest forward transfer on the final task: the T5 diagonal entry is 0.79, compared to 0.77 (iFOPNG), 0.76 (FOPNG), and 0.78 (EWC). The average accuracy gap between Adam and projection methods in the main text therefore reflects weaker forward transfer to later tasks, not stronger forgetting of earlier ones — a distinction visible directly in this matrix.

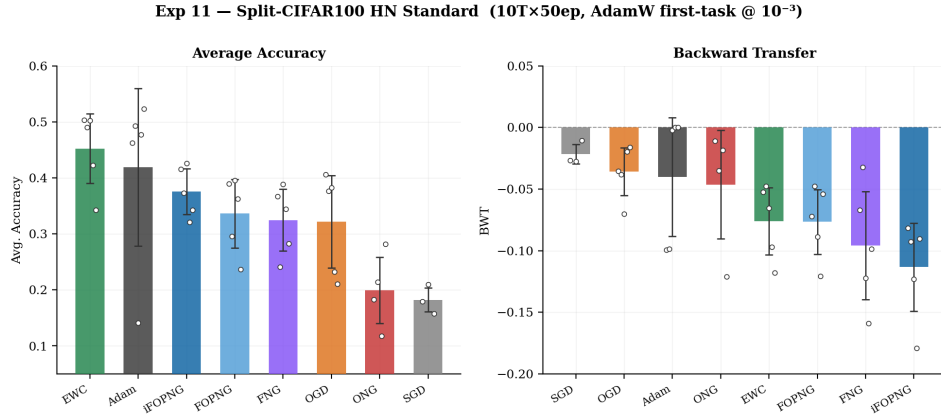


Figure 14: **Split-CIFAR100 hypernetwork — 10 tasks, 50 epochs per task, AdamW first-task initialisation at 10^{-3}** . Left: average accuracy. Right: backward transfer. Random-chance performance is 10% (10-way classification per task). EWC leads at 45.5%, with Adam second at 42.0% and iFOPNG third at 37.8% — a rank ordering that inverts the five-task Split-CIFAR10 pattern (where projection methods outperform EWC and closely trail Adam). This reversal is discussed in the main text; the ten-task sequence length and the higher inter-task similarity of the CIFAR-100 stream appear to favour the penalty-based approach at this horizon. Seed-level variance is elevated across all methods; notably, Adam contains one outlier seed at $\approx 14\%$, pulling the mean below the median. SGD (18.7%) and ONG (20.1%) barely exceed random chance, indicating near-complete forgetting on this benchmark.

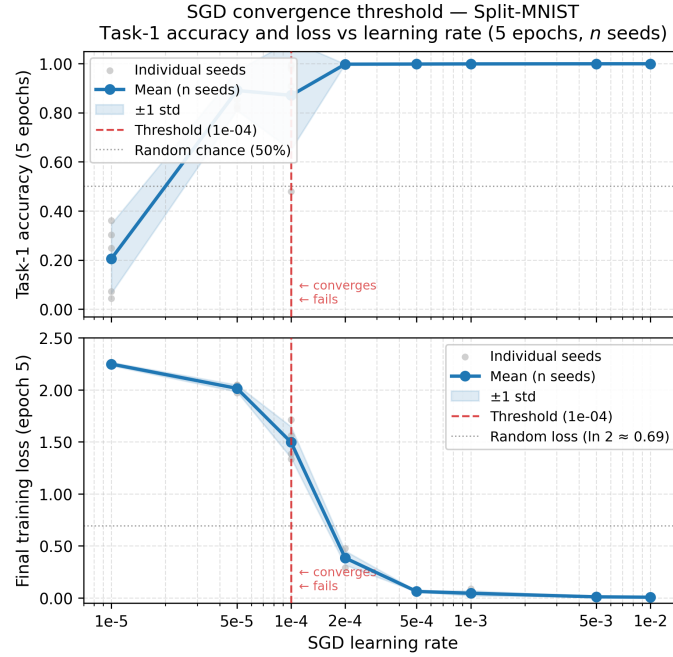


Figure 15: **SGD first-task convergence threshold — Split-MNIST (5 epochs, task-1 only).** Top: task-1 accuracy after five epochs of SGD training as a function of learning rate. Bottom: corresponding final training loss. The red dashed line at $\eta = 10^{-4}$ marks the empirical convergence threshold: below it, accuracy is at or near random chance (50%) and loss remains above 1.5; above it, the network converges reliably to near-perfect task-1 accuracy. The FOPNG paper specifies $\eta = 10^{-5}$ for first-task SGD training, which this sweep shows yields only $\approx 21\%$ accuracy — barely above random chance. The convergence threshold lies between 5×10^{-5} and 10^{-4} . All standalone experiments in this thesis use $\eta = 10^{-3}$, which achieves near-perfect first-task convergence and provides a clean initial projection subspace.

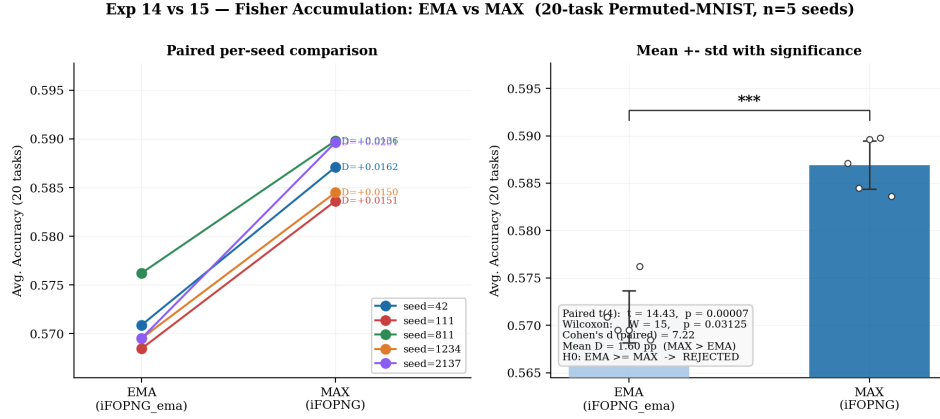


Figure 16: **EMA vs. MAX Fisher accumulation — per-seed statistical comparison, 20-task Permuted-MNIST, $n = 5$ seeds (Sub-RQ4).** Left: paired slope plot. Each line connects the final average accuracy of one seed under EMA (left) and MAX (right); the annotated D values give the per-seed difference. Right: mean $\pm$ std bar chart with the paired t -test significance bracket. Every seed shows a positive slope (MAX > EMA), confirming that the direction of the effect is consistent without exception. The per-seed differences range from +1.50 to +1.65 pp. Formal tests: paired $t(4) = 14.43$, $p < 0.001$; Wilcoxon signed-rank $W = 15$, $p = 0.031$; Cohen's $d = 7.22$. The null hypothesis that EMA performs at least as well as MAX is rejected by both tests. As discussed in the main text, the effect is statistically unambiguous but practically negligible (1.60 pp mean difference on a benchmark where both strategies lose more than 30 pp over the full sequence).

APPENDIX E: FULL BENCHMARK METHOD COMPARISON

This appendix provides a complete accuracy reference for all eight methods across every benchmark evaluated in this thesis. Figures 17 and 18 present side-by-side grouped bar charts contrasting standalone and hypernetwork performance on Split-CIFAR10 and Split-MNIST respectively, allowing direct reading of each method's accuracy and BWT under both settings. Table 1 collects mean average accuracy across all seven benchmarks in a single table for quick cross-benchmark reference. Following standard practice in the continual learning literature, results are reported as mean $\pm$ standard deviation over seeds without formal pairwise significance tests; formal statistical comparison is reserved for the targeted iFOPNG vs. FOPNG analysis in Sub-RQ3 (Section 5.3) and the EMA vs. MAX analysis in Sub-RQ4 (Section 5.4).

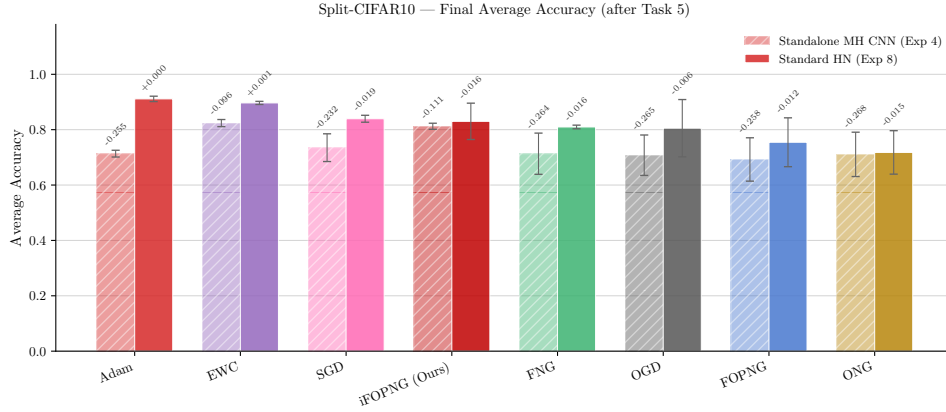


Figure 17: **Split-CIFAR10 — all methods, standalone MH CNN (left) vs. standard hypernetwork (right).** Bars show mean final average accuracy after all five tasks; BWT is annotated on each bar. The left panel corresponds to the standalone benchmark in Appendix B (Figure 8); the right panel corresponds to the main-text hypernetwork experiment (Figure 1). On the standalone benchmark EWC and iFOPNG lead (81.9% and 81.1%), while FOPNG is the weakest projection method (67.4%). In the hypernetwork setting the ranking partially reorders: Adam extends its lead (90.1%) and iFOPNG remains the best projection method (85.6%), but ONG drops substantially relative to its standalone position. The BWT annotations confirm that the hypernetwork’s functional regularizer suppresses forgetting across all methods relative to their standalone counterparts.

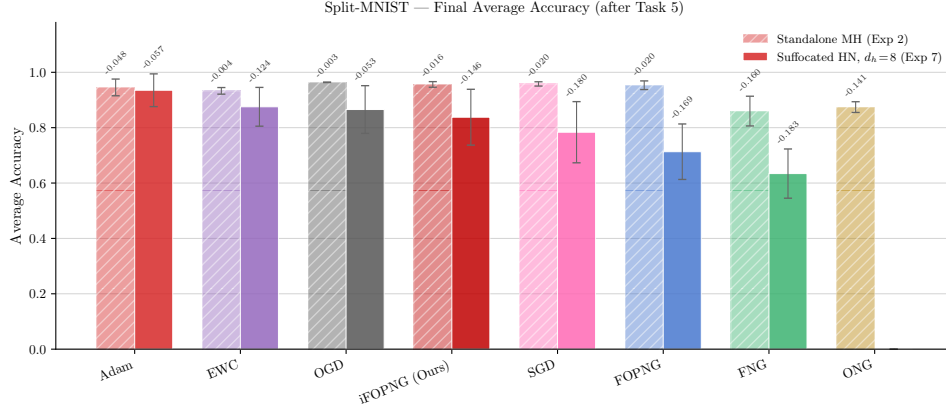


Figure 18: **Split-MNIST — all methods, standalone MH CNN (left) vs. suffocated hypernetwork at $d_h = 8$ (right)**. Bars show mean final average accuracy after all five tasks; BWT is annotated on each bar. The left panel corresponds to Appendix B (Figure 7, bottom row); the right panel corresponds to the main-text suffocated-regime experiment (Figure 2). On the standalone benchmark most methods cluster near 96%, with ONG and FNG as notable exceptions. Under extreme compression ($d_h = 8$) the ranking restructures substantially: Adam maintains high accuracy (93.4%) while Fisher-based projection methods (iFOPNG, FOPNG, FNG) collapse to 55–70% despite near-zero BWT, revealing the plasticity failure described in the main text. OGD is the most resilient projection method (86.6%), consistent with its Fisher-free projection mechanism. Note that several methods completed fewer than three seeds in the suffocated regime; bars with low seed counts should be interpreted with caution.

Table 1: **Mean average accuracy (%) across all benchmarks and methods**. Values are seed-averaged final average accuracy after all tasks. Standalone benchmarks use multi-head target networks; HN benchmarks use the chunked hypernetwork architecture. [†]MNIST HN is the suffocated regime ($d_h = 8$). [‡]CIFAR-100 MH EWC used $\lambda = 10$; $\lambda = 50$ collapsed. **Bold**: best per column. ***Bold italic***: iFOPNG. —: insufficient seeds or unavailable.

Method	Standalone				Hypernetwork		
	Perm-MNIST	MNIST MH	CIFAR-10 MH	CIFAR-100 MH [‡]	MNIST HN ($d_h=8$) [†]	CIFAR-10 HN	CIFAR-100 HN
<i>iFOPNG</i> (ours)	86.1	95.9	81.2	42.7	69.9	85.6	37.8
FOPNG	85.8	95.3	67.4	37.1	59.9	77.5	33.8
EWC	83.8	93.2	81.9	46.8	84.5	81.5	45.5
OGD	80.8	96.1	71.4	49.2	86.6	78.3	32.3
FNG	69.2	86.1	70.0	37.5	54.8	80.9	32.5
ONG	63.0	87.1	68.8	32.3	—	67.2	20.1
SGD	84.9	96.1	68.9	45.4	79.8	84.1	18.7
Adam	68.2	95.1	68.1	40.5	93.4	90.1	42.0
Random chance	50.0	50.0	50.0	10.0	50.0	50.0	10.0

APPENDIX F: HYPERPARAMETER TABLES

All values are sourced directly from the W&B run configs (CSV logs) unless marked [?] (inferred from thesis text or figure annotations). Entries marked [†] are flagged in Section 8. `max_ep` is the early-stopping patience; `ep/task` is epochs per task; `max_dirs` is the gradient memory column budget. Values that are constant across all experiments in a group are stated in the relevant caption rather than repeated in every row.

F.1 Standalone Target-Network Experiments

Table 2: **Standalone training hyperparameters.** All rows: `model = TargetNetwork`, `normalize = False`, `regulizer = False`, `damping = 0.01`, $n = 5$ seeds.

Benchmark	Tasks	ep/task	max_ep	First opt	First LR	LR	Batch	Fisher samp.
Permuted-MNIST	5	10	5	SGD	10^{-3}	10^{-4}	10	60 000
Split-MNIST MH	5	10	5	SGD	10^{-5}	10^{-5}	10	12 000
Split-CIFAR10 MH	5	10	5	Adam	10^{-3}	10^{-3}	10	1 024
Split-CIFAR100 MH	10	10	10	SGD	10^{-2}	5×10^{-3}	10	1 024

Table 3: **EWC λ for standalone experiments.** Projection methods and Adam/SGD use $\lambda = 0.001$ (minimal). [†]CIFAR-100 MH: $\lambda = 50$ collapsed and is excluded; $\lambda = 10$ is used (annotated in Figure 9). [?]Other EWC values inferred from the set of distinct `lam` values in each experiment’s logs.

Benchmark	EWC λ	Log values present
Permuted-MNIST	$10^?$	{10, 0.01, 0.001}
Split-MNIST MH	$0.0005^?$	{400, 0.001, 0.0005}
Split-CIFAR10 MH	10 or $50^?$	{100 000, 50, 10, 0.001}
Split-CIFAR100 MH	$10^{\dagger}$	Confirmed; $\lambda = 50$ excluded

F.2 Hypernetwork Experiments

All HN experiments: `model = HyperNetwork`, `regulizer = True`, $\beta = 0.001$ (`lam = 0.001`), $\eta = 10^{-3}$, $n = 5$ seeds.

Table 4: **Hypernetwork architecture and training parameters.** `c_emb` = chunk embedding dim; `t_emb` = task embedding dim. [†]Exp 407: `first_task_opt` = None in logs; likely AdamW by code default (see F.6). [†]Exp 411: extracted from the 35 completed runs at chunk = 6 000, $d_h = 32$, `normalize` = True.

Role / Benchmark	Tasks	ep/task	max_ep	First opt	d_h	chunk	c_emb	t_emb	grads / task
CIFAR-10 HN (main)	5	10	50	AdamW	32	6 000	16	16	80
CIFAR-100 HN [†]	10	10	50	AdamW	32	6 000	16	16	80
MNIST HN ($d_h = 8$, suffocated) [†]	5	10	15	AdamW	8	64	32	32	100
dh sweep $d_h = 4$	5	10	15	AdamW	4	64	32	32	80
dh sweep $d_h = 8$	5	10	15	AdamW	8	64	32	32	100
dh sweep $d_h = 16$	5	10	15	AdamW	16	64	32	32	80

F.3 Normalization Ablation

All three conditions share the same architecture: Split-CIFAR10, 5 tasks, `max_ep` = 50, chunk = 6 000, $d_h = 32$, batch = 32, AdamW first-task at $\eta = 10^{-3}$, iFOPNG only.

Table 5: **Normalization ablation conditions (Sub-RQ2).** The CSVs for conditions 1 and 2 also contain pilot runs at chunk = 256, $d_h = 64$; these are excluded. Only chunk = 6 000, $d_h = 32$ runs are used, confirmed by the reported accuracy values.

Condition	normalize	QR on G	Max-scale $\hat{F}$	Seeds	Accuracy	BWT
No normalization	False	×	×	3	72.2%	−0.070
Gradient-only	True	✓	×	3	68.9%	−0.060
Full normalization	True	✓	✓	3	85.8%	−0.013

F.4 Sub-RQ4: Fisher Accumulation Operators

Both conditions: 20-task Permuted-MNIST, TargetNetwork, `normalize` = False, batch = 10, $n = 5$ seeds.

Table 6: **Sub-RQ4 accumulation parameters.** **Note:** the Discussion incorrectly states $\alpha = 0.3$; the confirmed value from the run logs is $\alpha = 0.5$.

Operator	Tasks	ep/task	max_ep	LR	Fisher samp.
MAX (iFOPNG)	20	10	5	10^{-4}	60 000
EMA (iFOPNG_ema, $\alpha = 0.5$)	20	10	5	10^{-4}	60 000

F.5 SGD Convergence Threshold Diagnostic

The SGD threshold diagnostic (Figure 15) sweeps $\eta \in \{10^{-5}, 5 \times 10^{-5}, 10^{-4}, 2 \times 10^{-4}, 5 \times 10^{-4}, 10^{-3}, 5 \times 10^{-3}, 10^{-2}\}$ on Split-MNIST SH (TargetNetwork, 1 task, 5 epochs). Convergence threshold: $\eta = 10^{-4}$. All standalone experiments in this thesis use $\eta = 10^{-3}$.

F.6 Notes on Confounds and Uncertainties

EXP 407 `first_task_opt = none`. The run config records `first_task_opt = None` and `first_task_lr = None`. The figure title annotates “AdamW first-task,” likely reflecting a code-level default: Exps 412 and 413 (same architecture, added later) explicitly log `adamw`, suggesting Exp 407 predates that config field.

EXP 411 PARAMETER HETEROGENEITY. The CIFAR-100 HN CSV contains 222 runs across a wide range of `chunk_size`, `hyper_hidden_dim`, and normalization settings — an exploratory search prior to settling on the published architecture. The 35 completed runs at `chunk=6000`, $d_h=32$, `normalize=True` are used for the reported results; all other runs are pilots.

NORMALIZATION ABLATION PILOT RUNS. Exps 409 and 410 each contain additional runs at `chunk=256`, $d_h=64$ (an earlier pilot) alongside the published `chunk=6000`, $d_h=32$ runs. The pilot runs reach near-random accuracy ($\approx 50\text{--}61\%$) and are excluded; the ablation is architecturally controlled.

EWC λ IDENTIFICATION. The `lam` field is shared between EWC’s λ and the projection damping term, so per-method EWC values cannot be extracted from the CSV without row-level method filtering; Table 3 values are inferred from the set of distinct `lam` values per experiment.